Journée de la recherche
Neural 3D Implicit Modeling

Karim Kassab[1,2], March 2025

Supervisors
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PhD Context

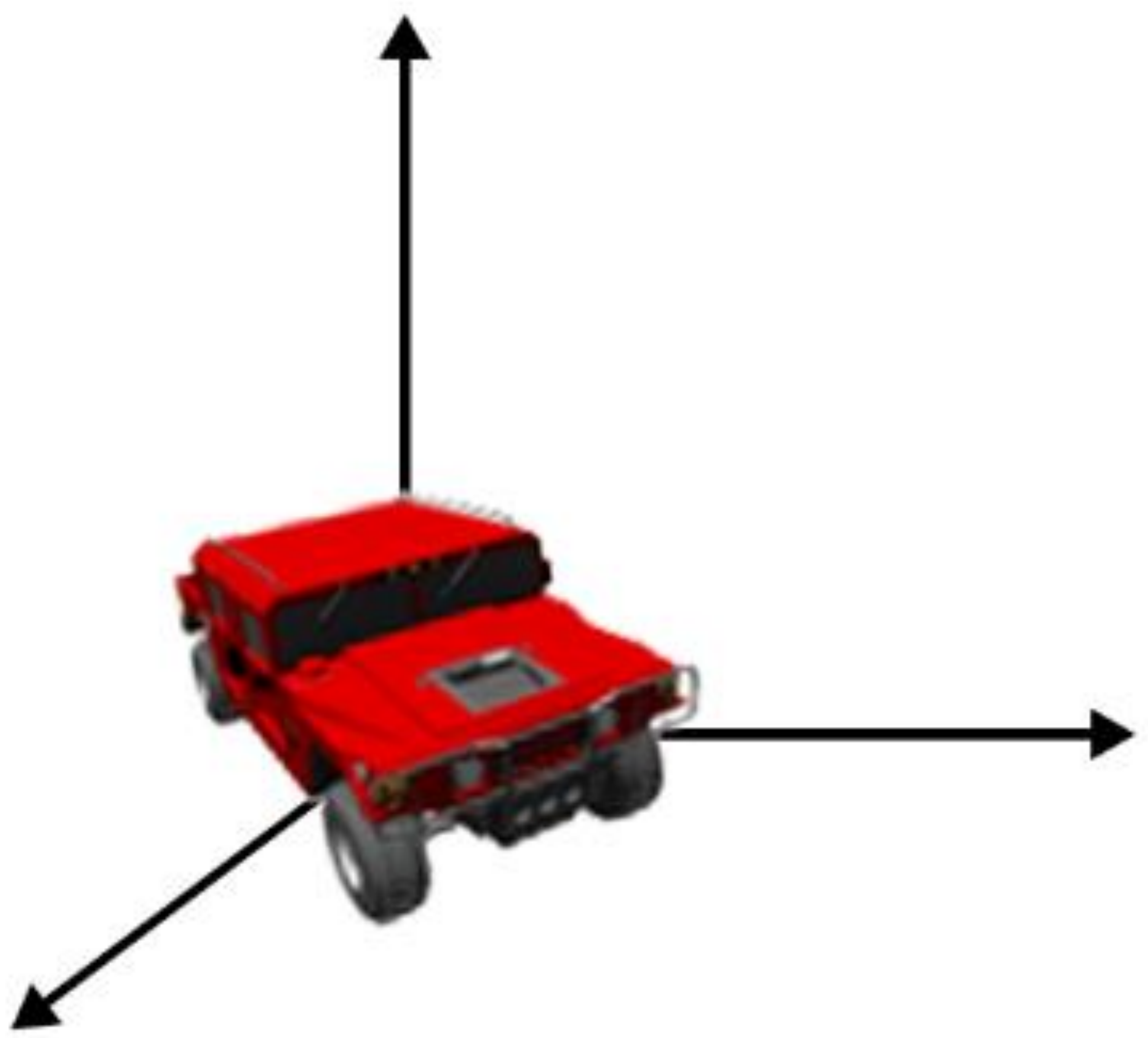
Start date: 30/01/2023

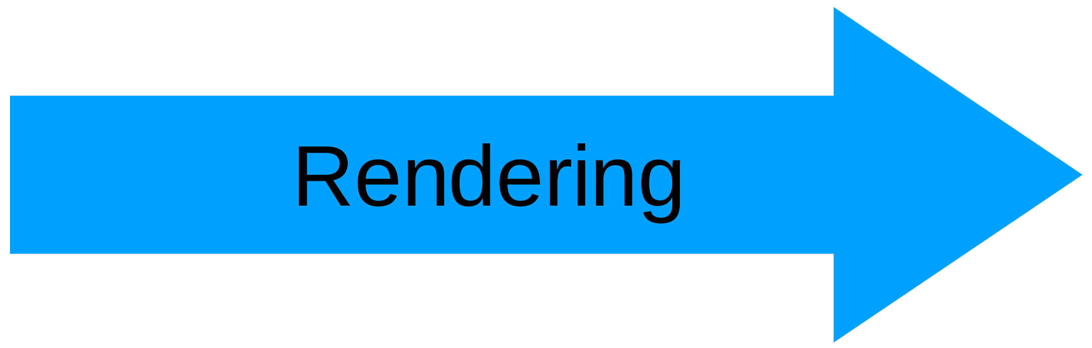
- **Key problem:** 3D modeling and asset generation
- **Premise:** Both Criteo and IGN own datasets of images that are taken in uncontrolled setups.
- How can we leverage 3D implicit models in such contexts?

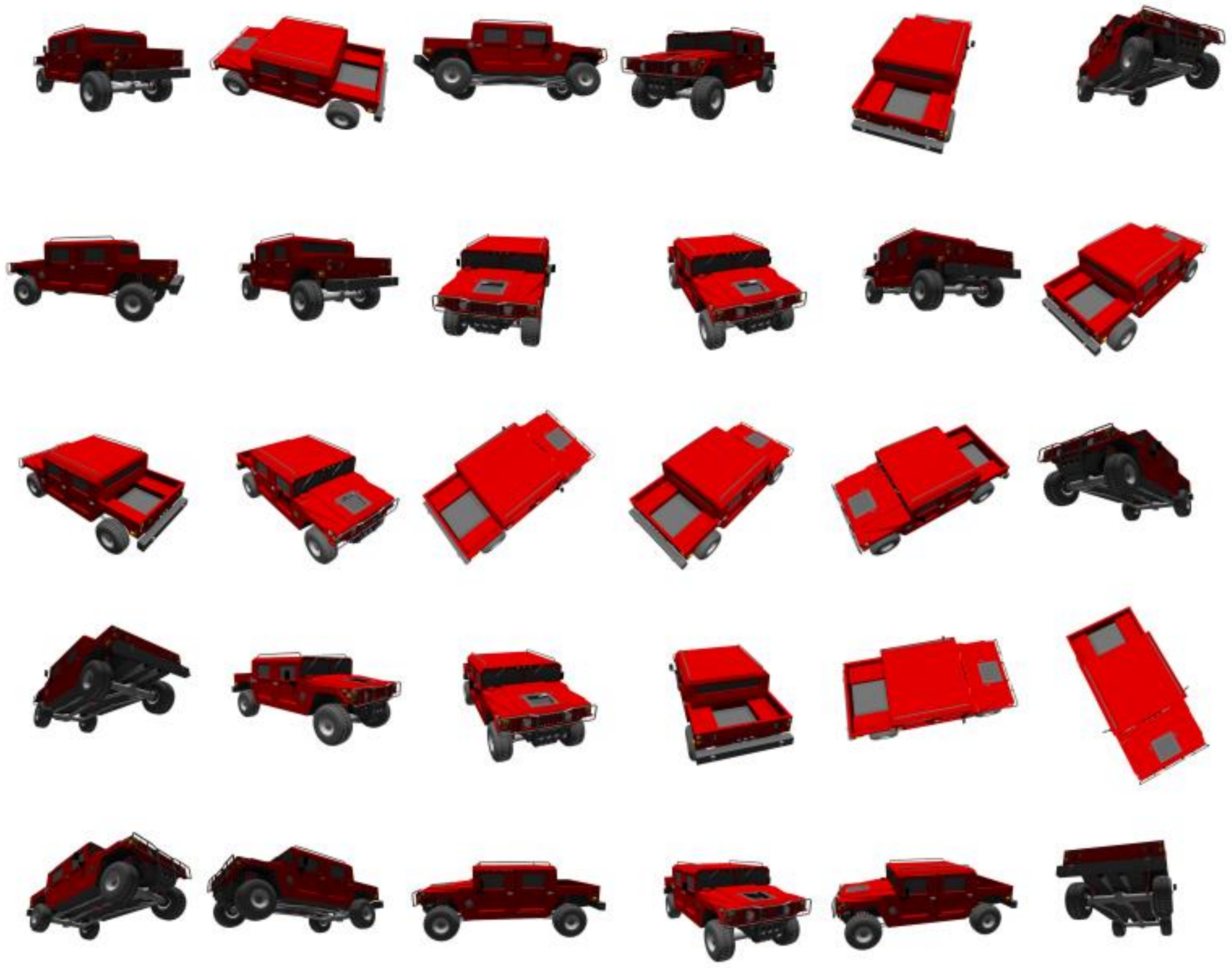
Inverse Graphics Problem

How to model a 3D object representation using its images?

Inverse Graphics Problem

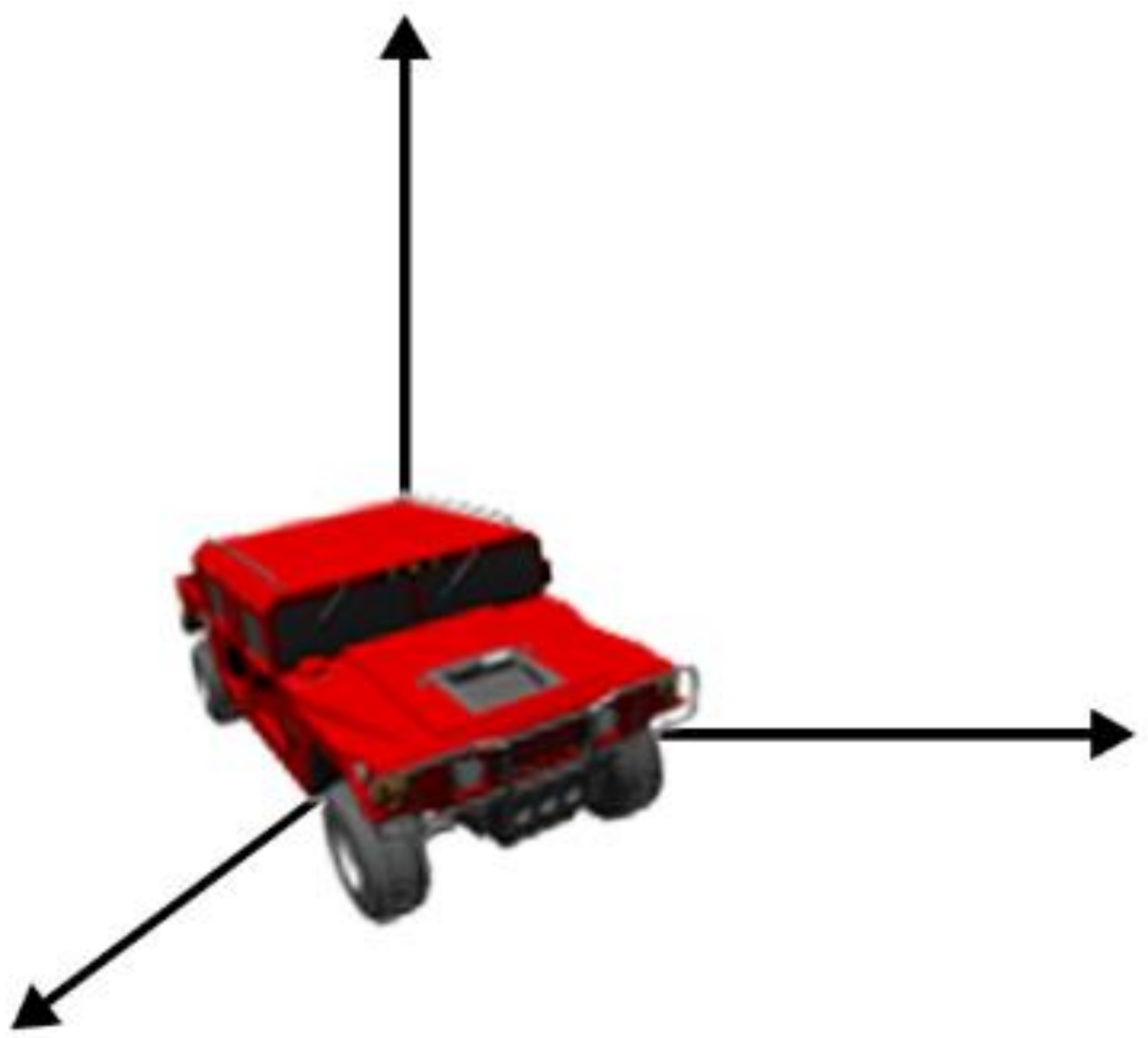






Mesh
Voxel grid
Point Cloud
SDF

Inverse Graphics Problem



Mesh
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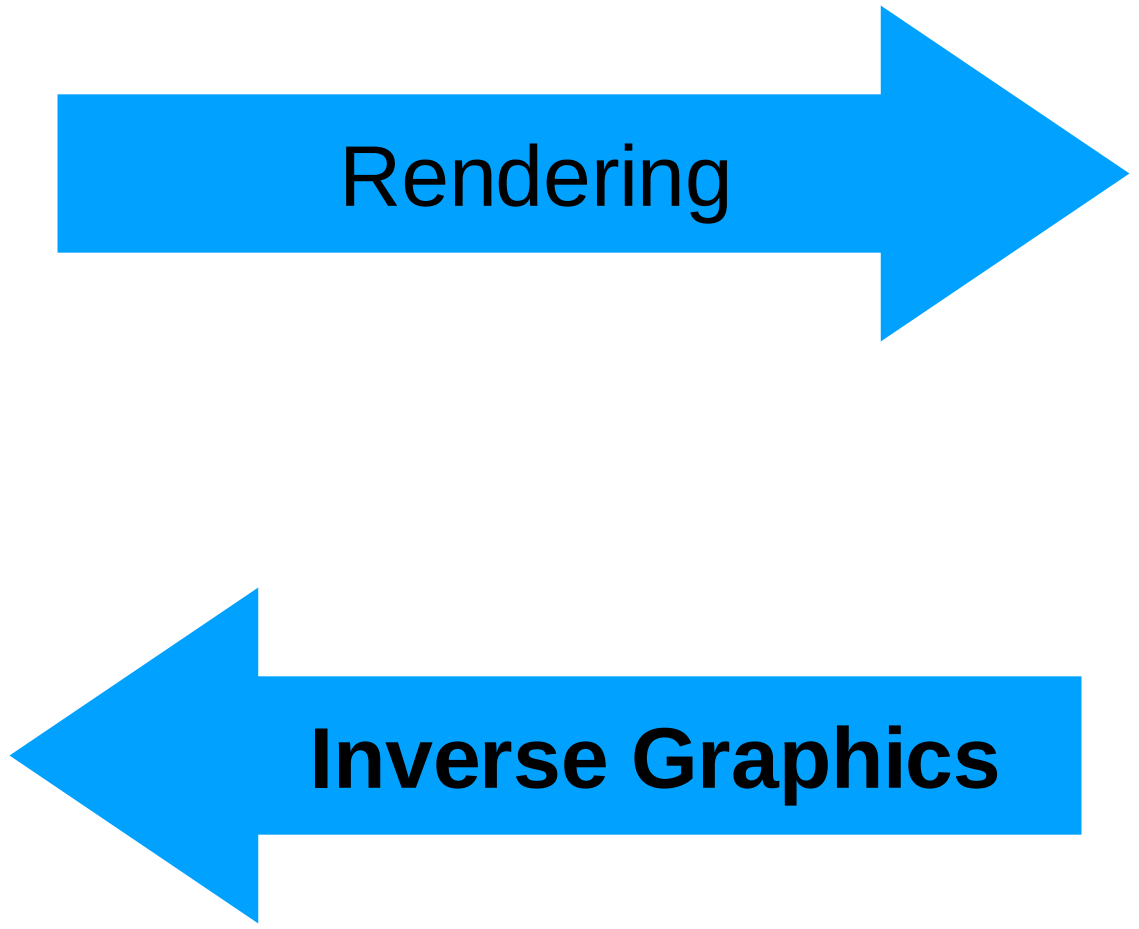
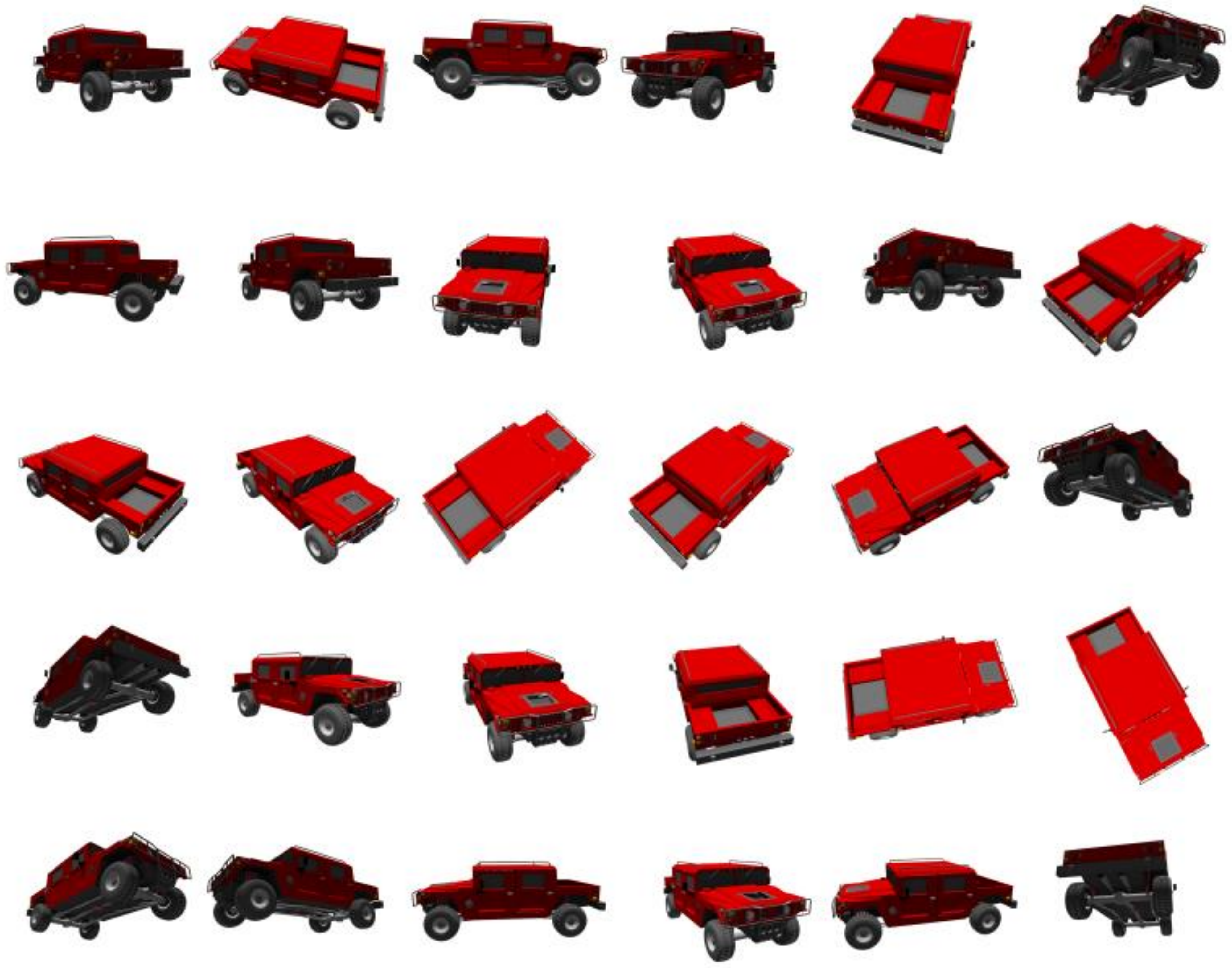
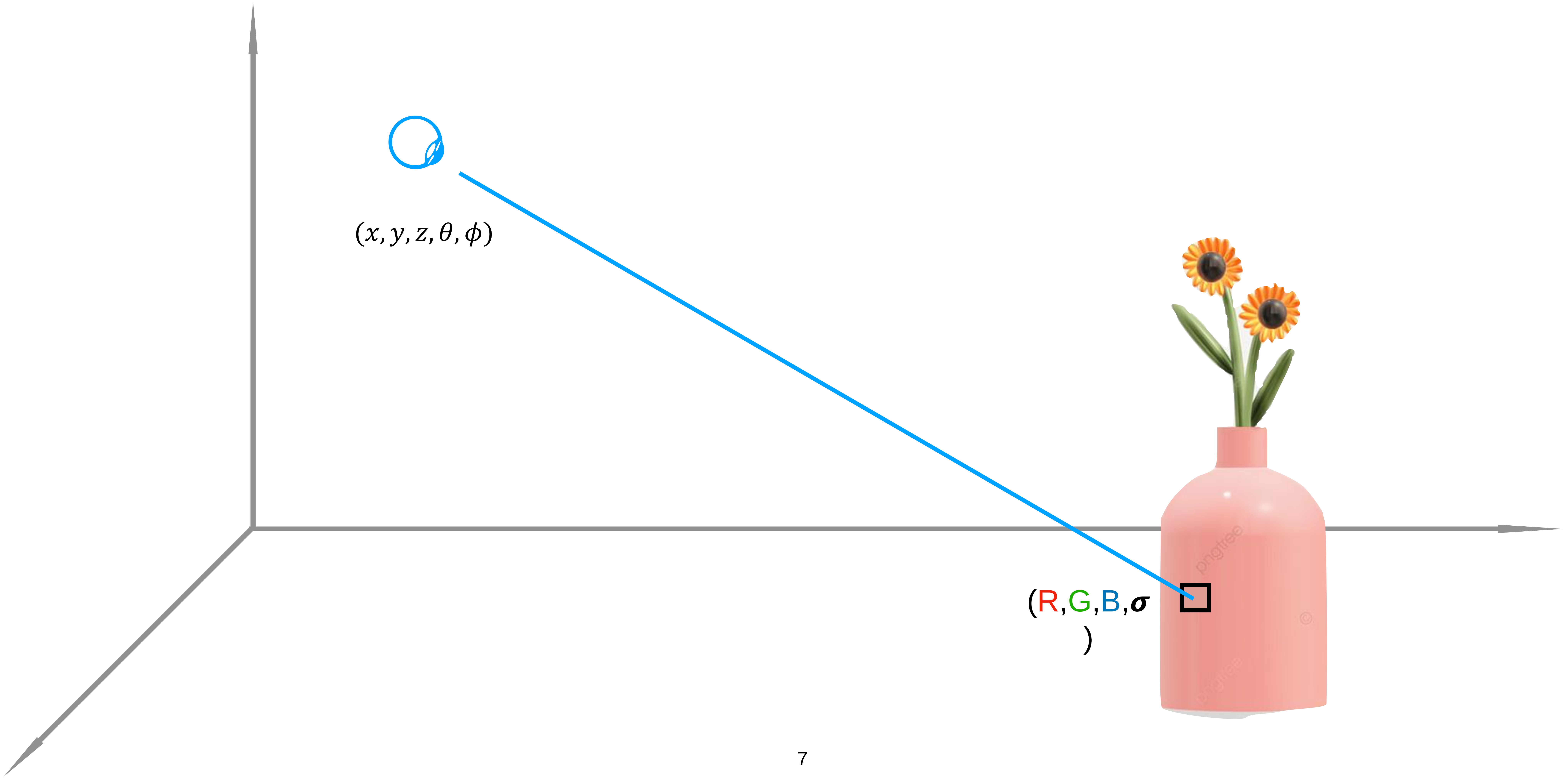




Neural Radiance Fields

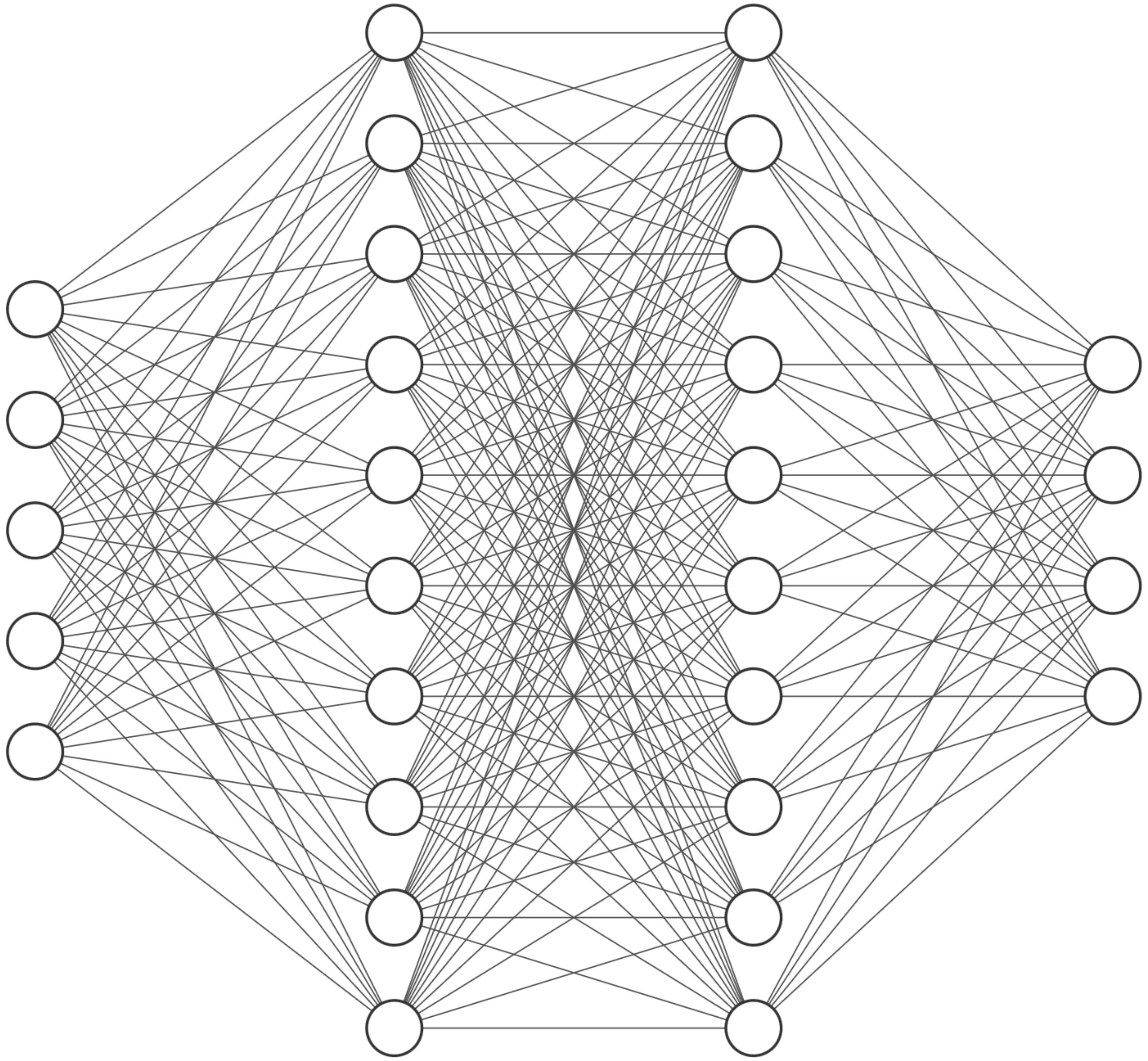
Neural Radiance Fields

$$f : (x, y, z, \theta, \phi) \longrightarrow (R, G, B, \boldsymbol{\sigma})$$



Neural Radiance Fields

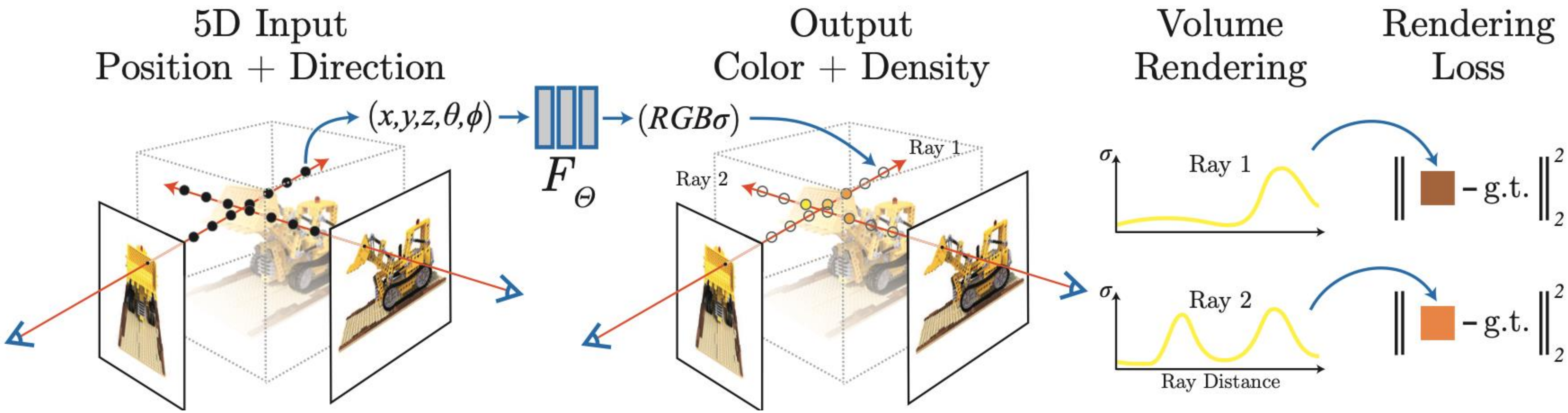
(x, y, z, θ, ϕ)



$(R, G, B, \boldsymbol{\sigma})$

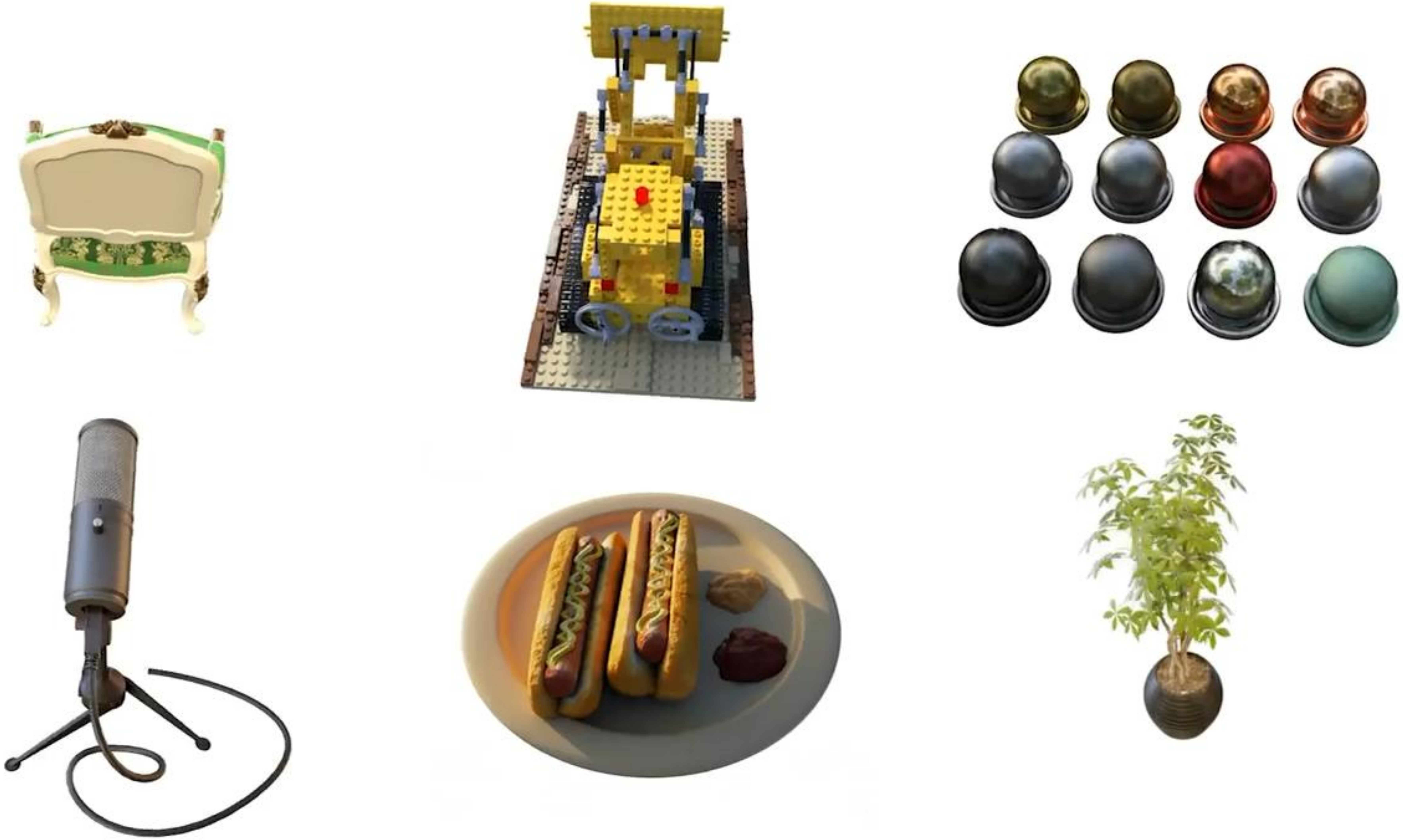
(Mildenhall et al., 2020)

Neural Radiance Fields



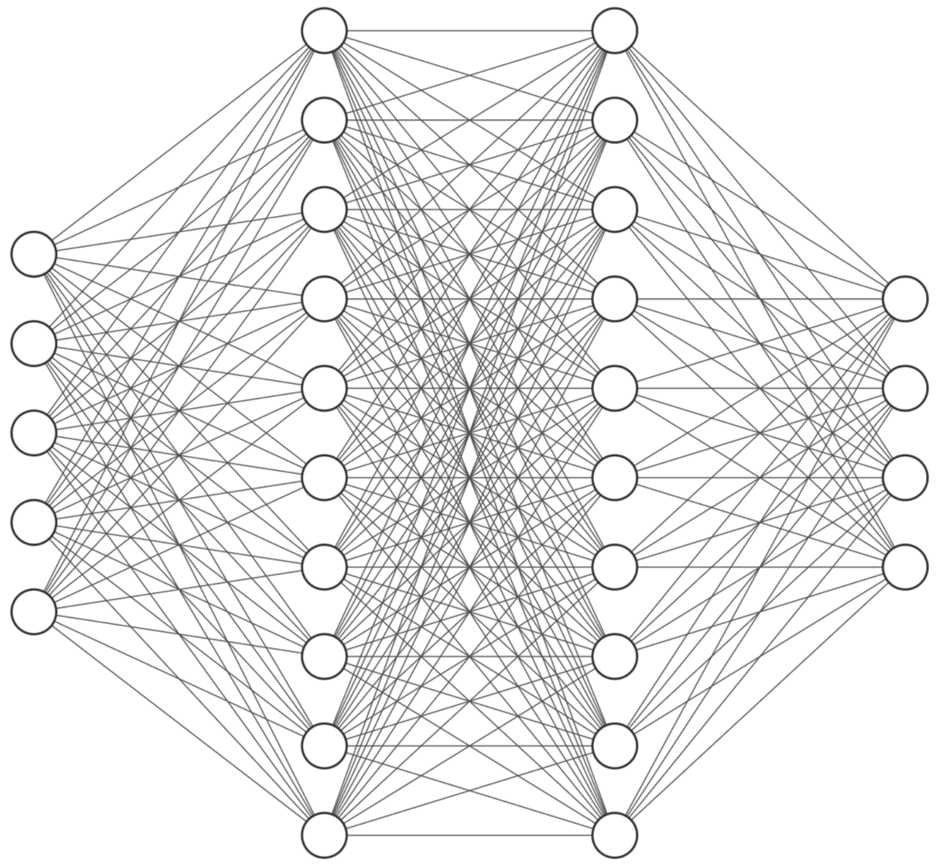
[NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis]

- **NeRFs are implicit representations trained to replicate views of a scene**
- **NeRF ≅ MLP + Volume Rendering Equations**



NeRF

(2020)

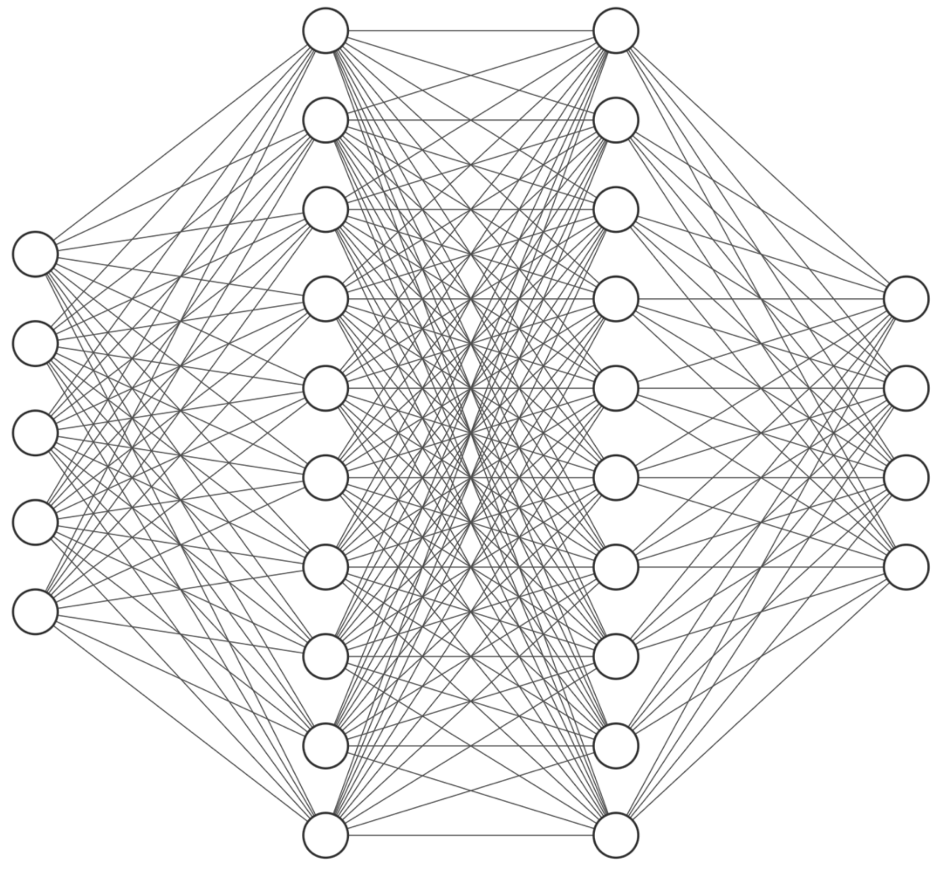


100% implicit

📉 **Memory**
📈 **Rendering time**

NeRF

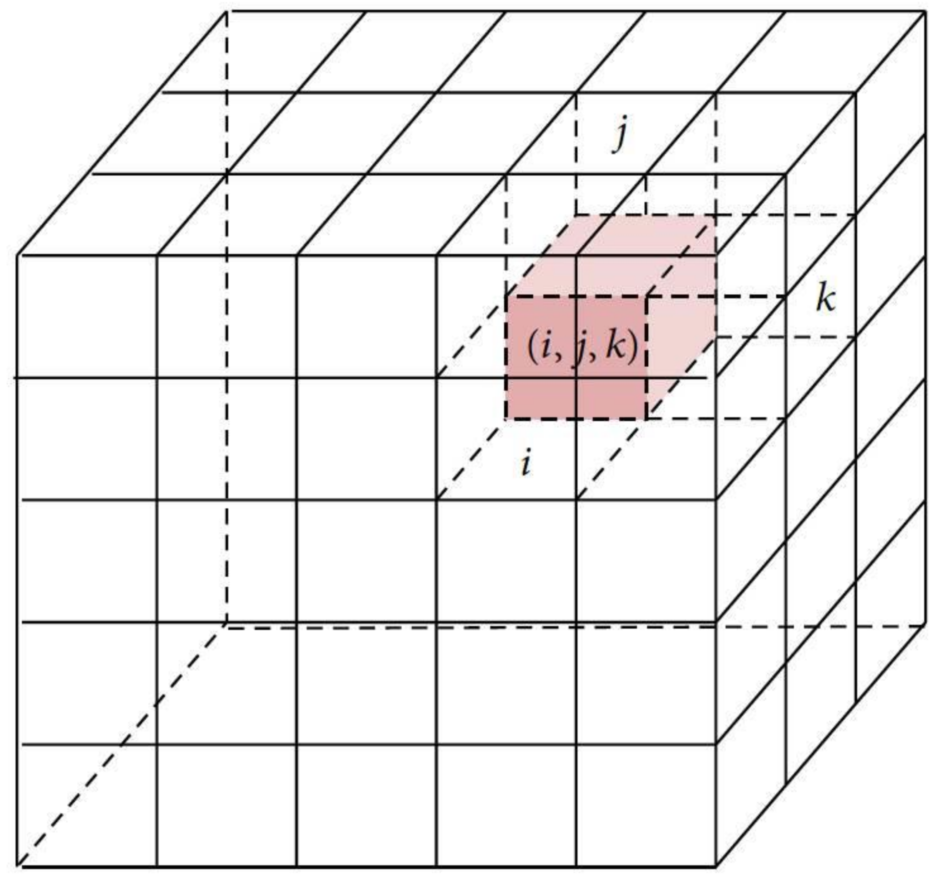
(2020)



100% implicit

📉 **Memory**
📈 **Rendering time**

Voxel Grids

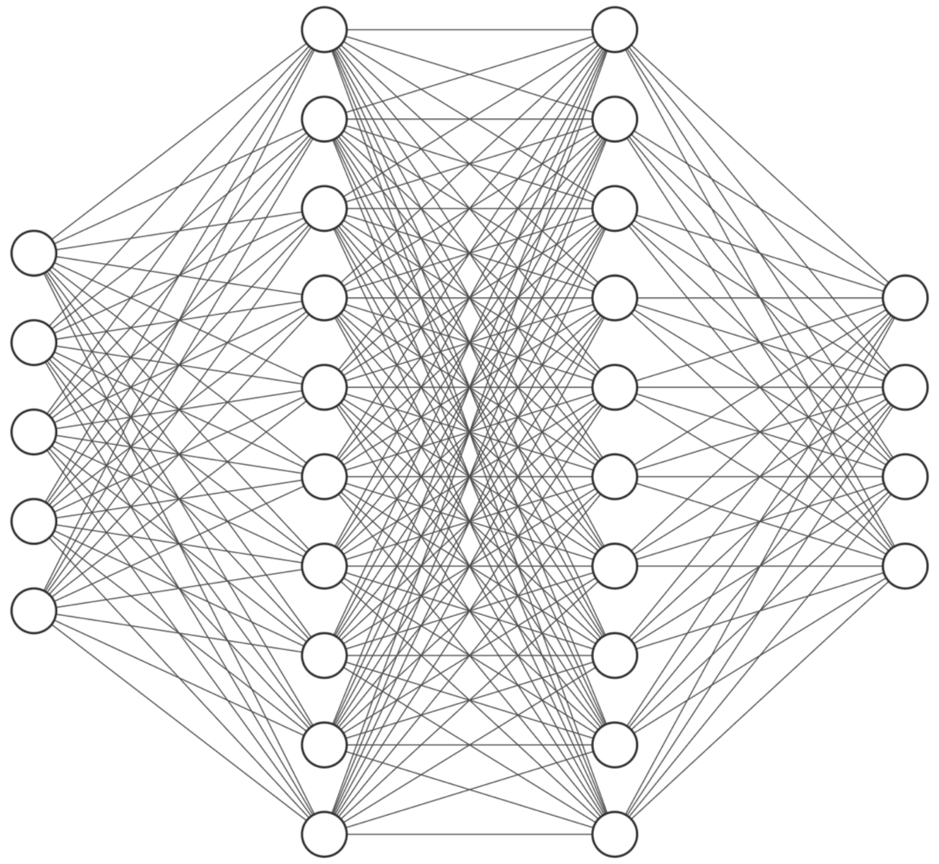


100% explicit

📈 **Memory**
📉 **Rendering time**

NeRF

(2020)

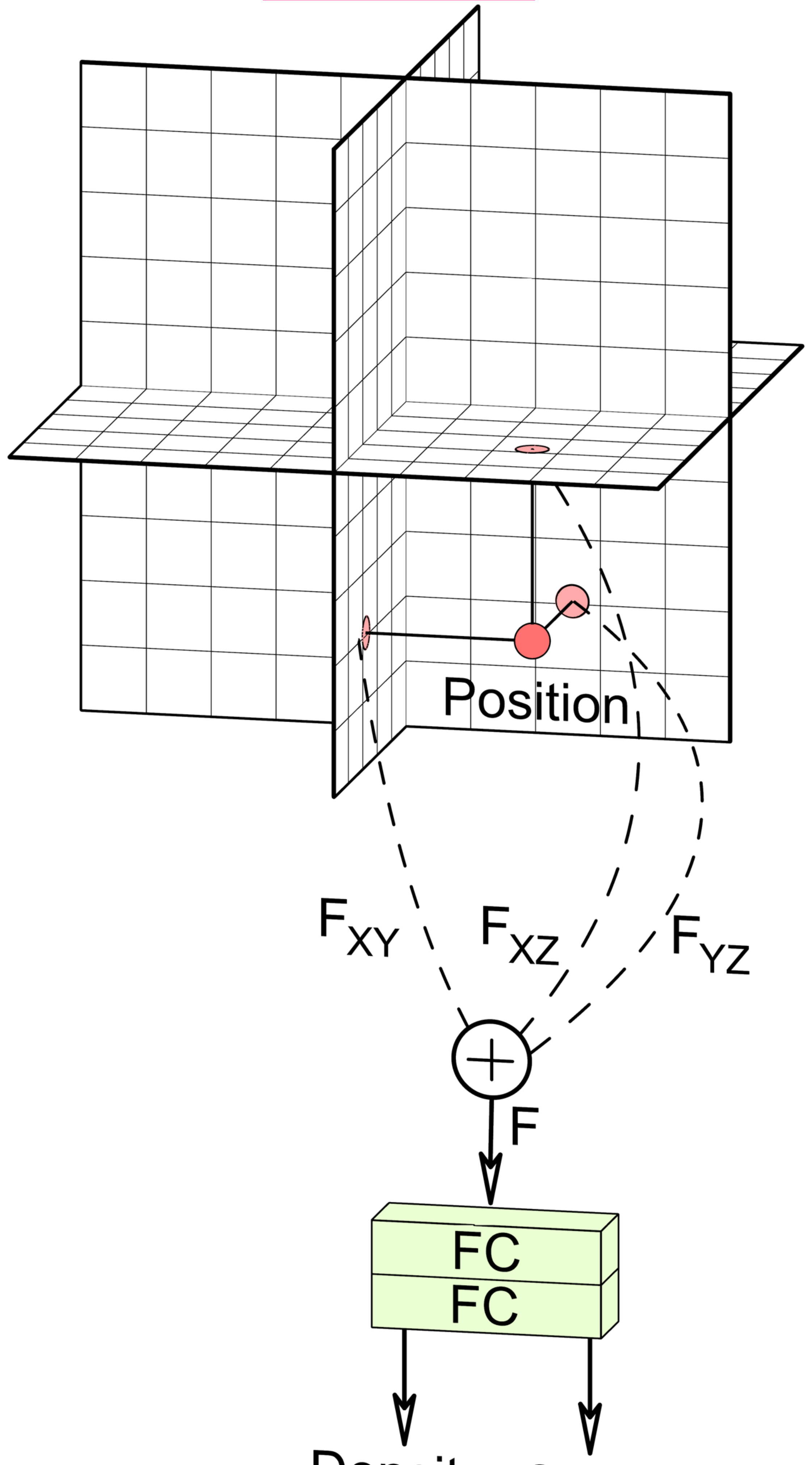


100% implicit

📉 Memory
📈 Rendering time

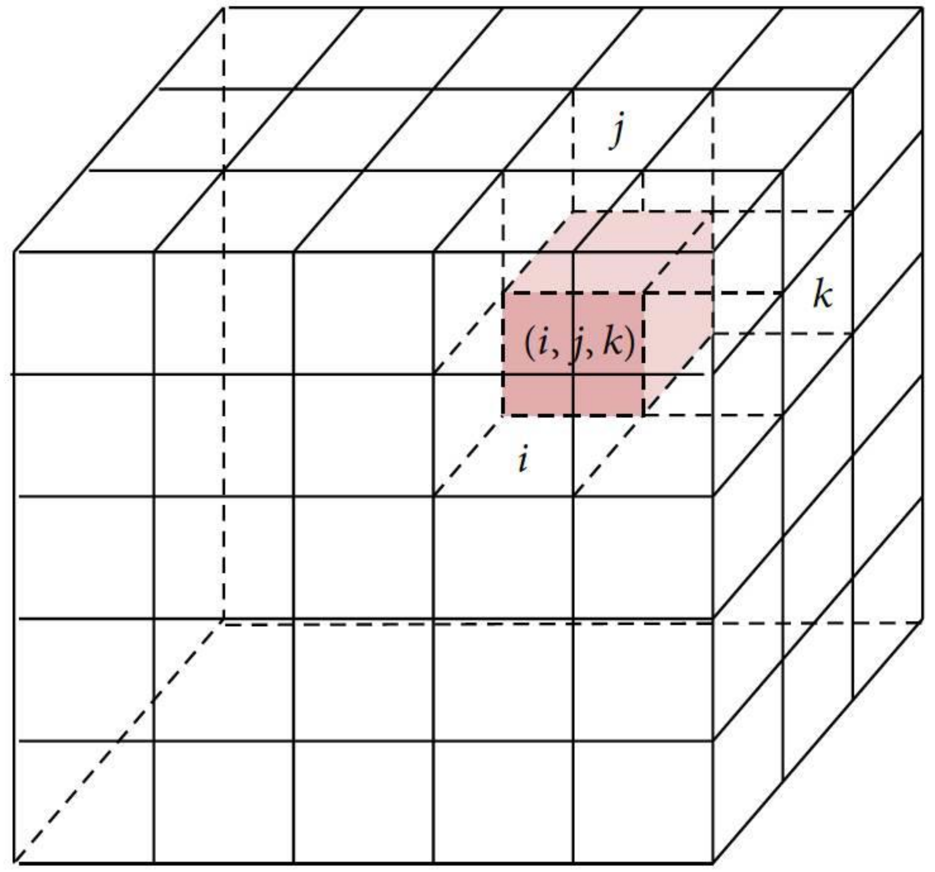
Tri-Planes

(2022)



Hybrid

Voxel Grids



100% explicit

📈 Memory
📉 Rendering time

Bringing NeRFs to the Latent Space: Inverse Graphics Autoencoder

Karim Kassab[*1,2], Antoine Schnepf[*1,3],
Jean-Yves Franceschi[1], Laurent Caraffa[2], Flavian Vasile[1], Jeremie Mary[1],
Andrew Comport[3], Valérie Gouet-Brunet[2]

Accepted at the International Conference on Learning Representations (ICLR) 2025

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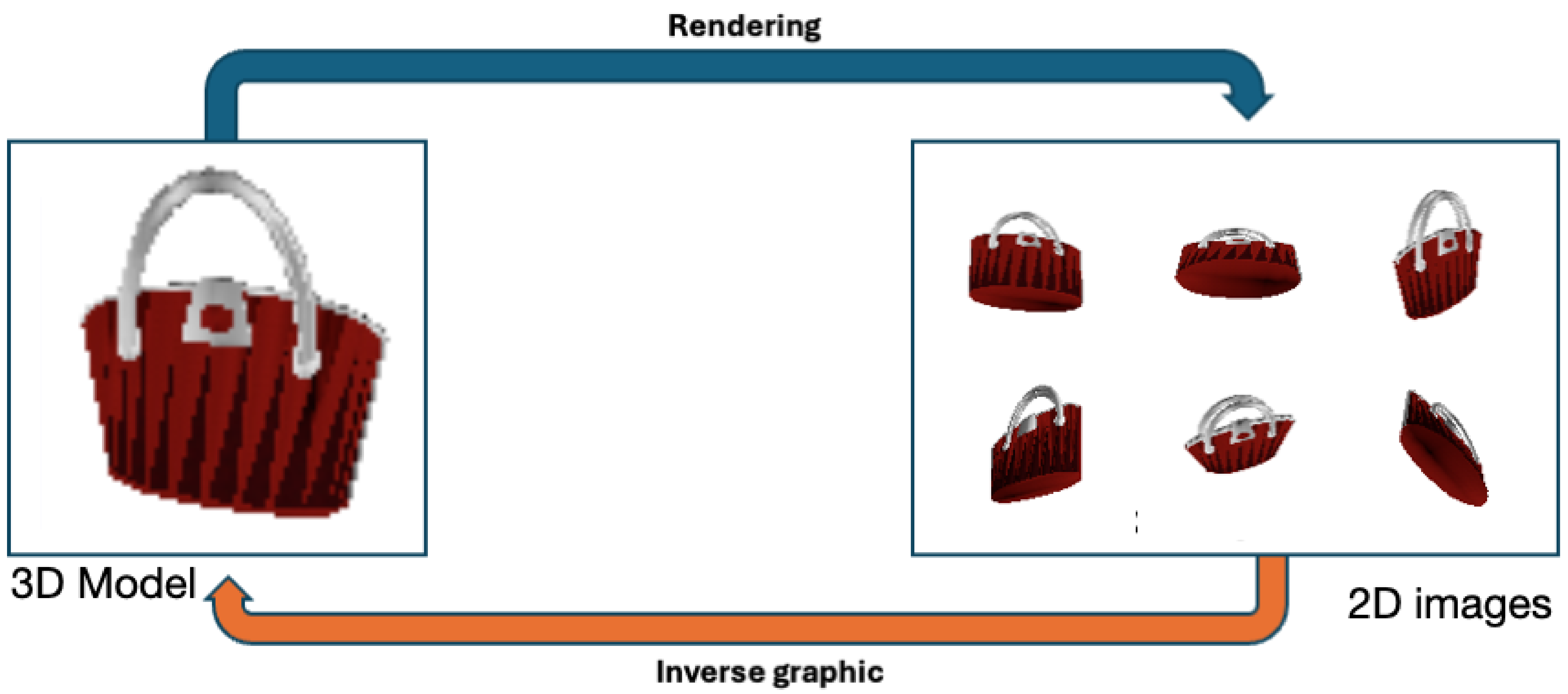
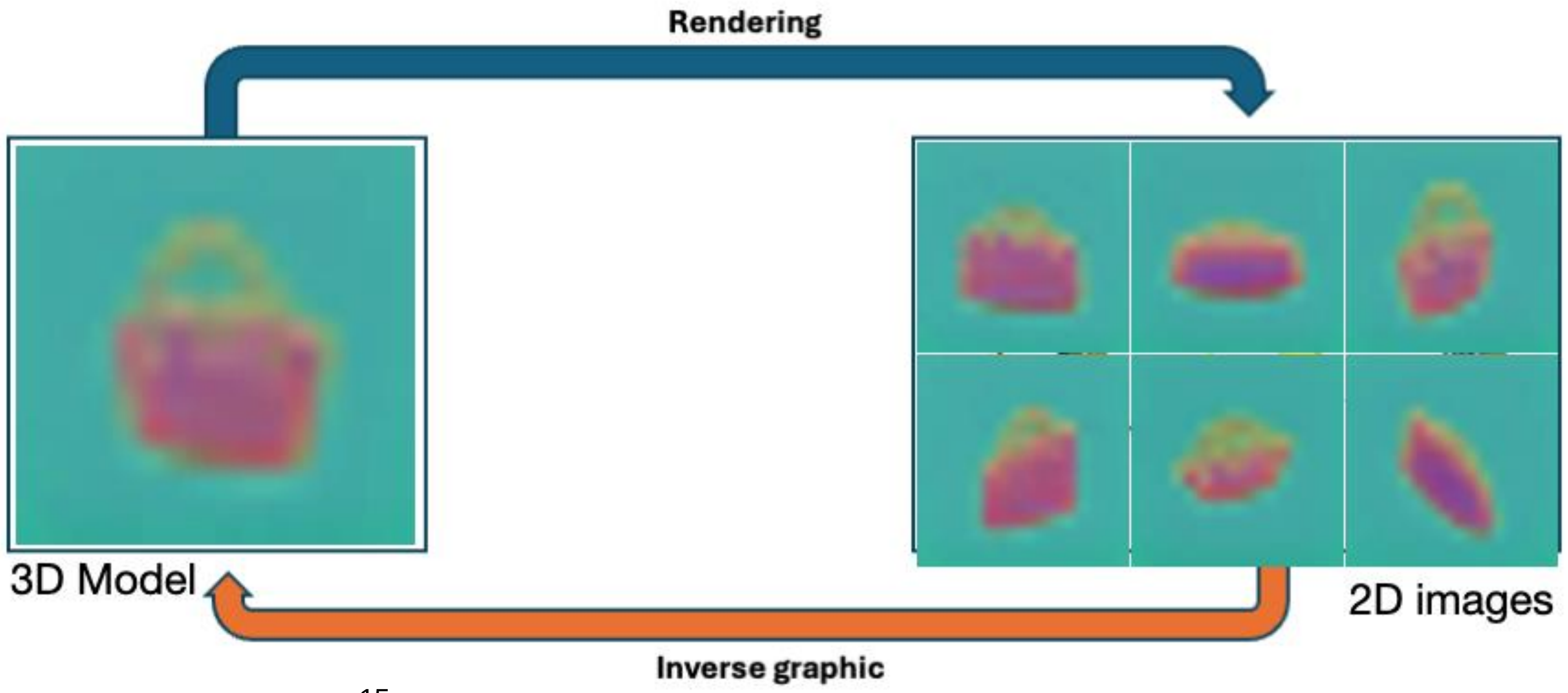
[3] Université Côte d'Azur, CNRS, I3S, France

ig-ae.github.io

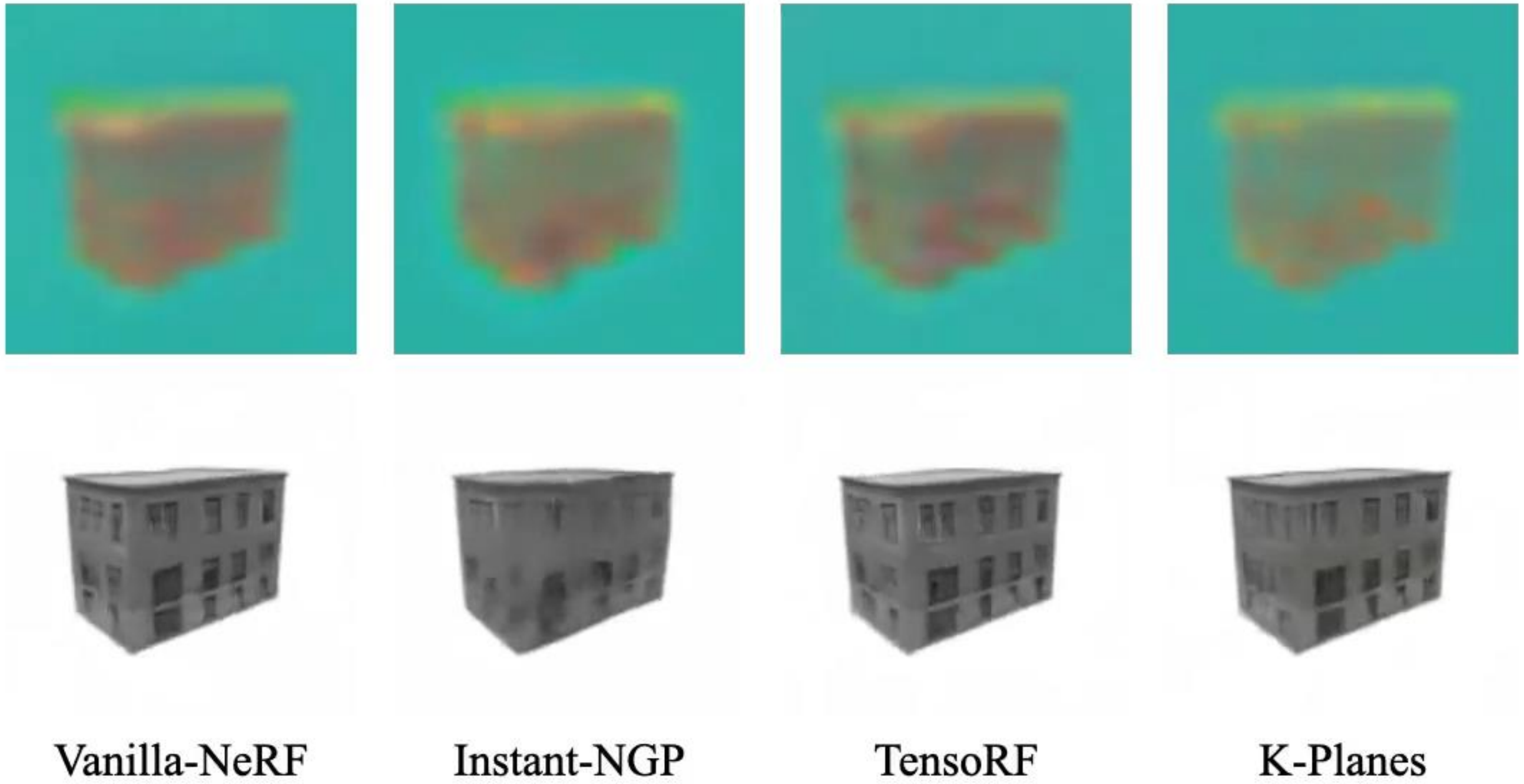
RGB NeRF



Latent NeRF



Latent NeRFs



Why is this interesting?

128x128

☹ Slow to render

16x16

😃 Fast to render

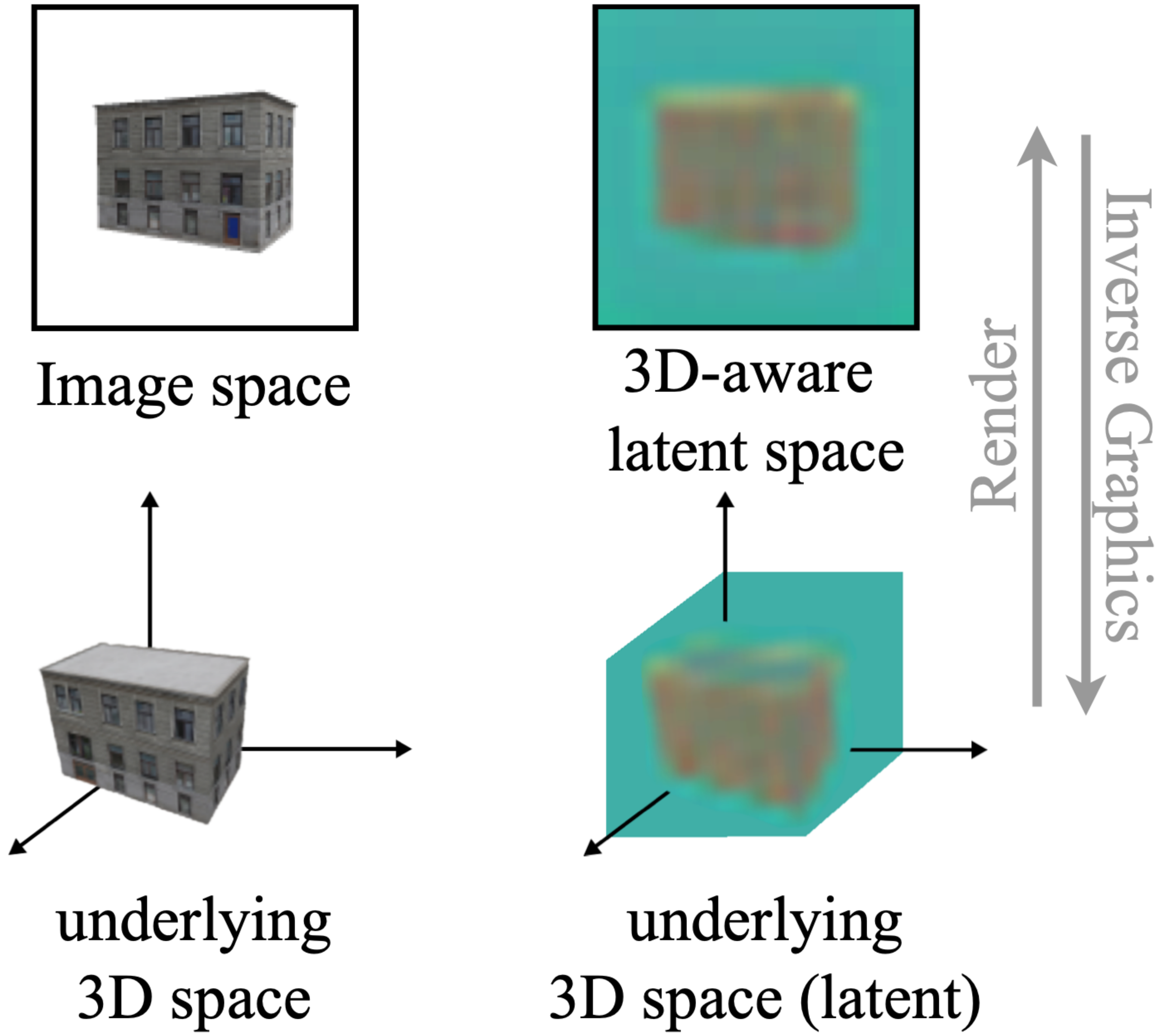
Why is this interesting?

😞 Slow to render

😃 Fast to render

Goal

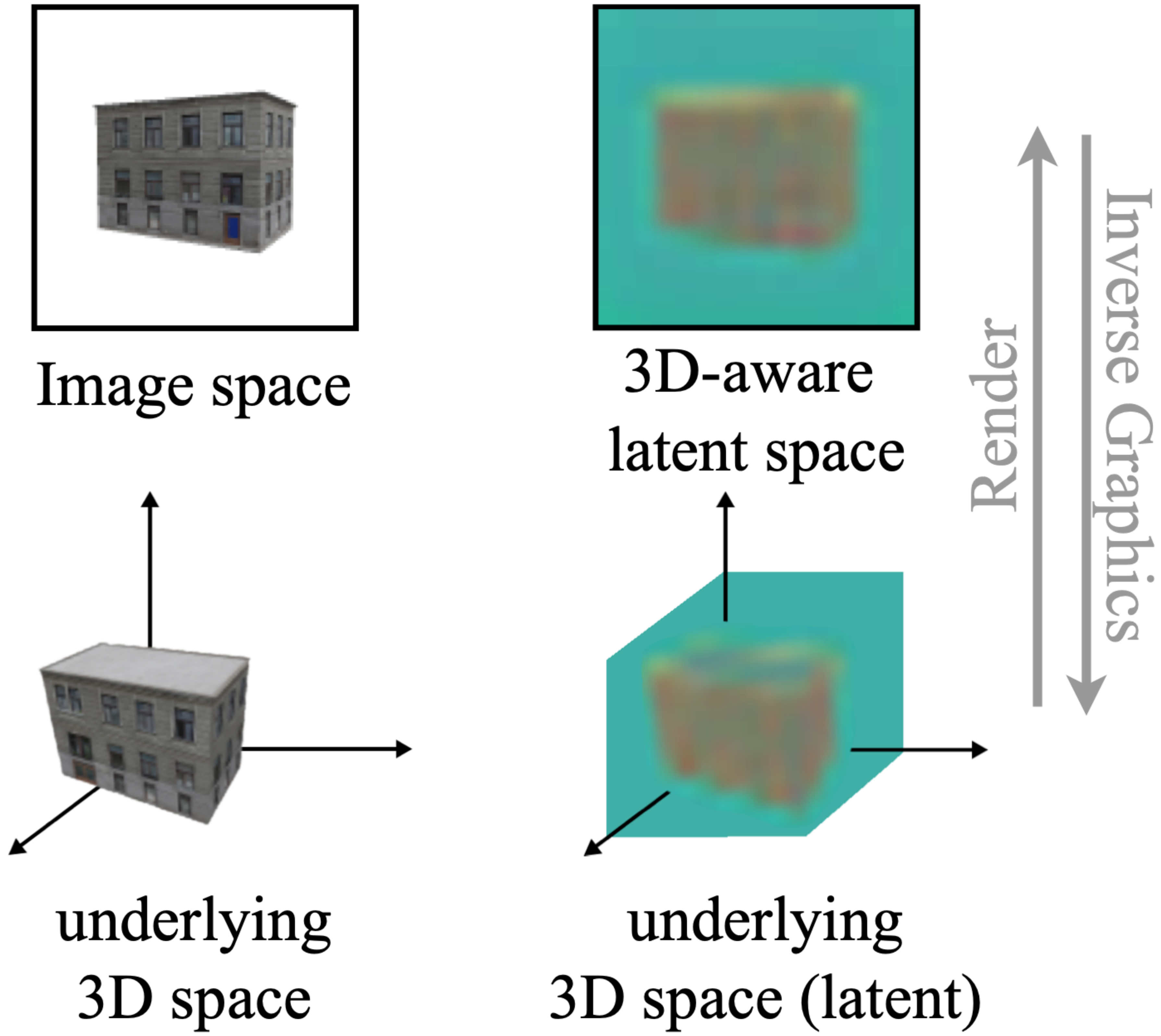
- Propose a latent NeRF training technique compatible with most NeRF architectures
- Build a 3D-aware latent space in which training NeRF is possible



(a) Image space analogy.

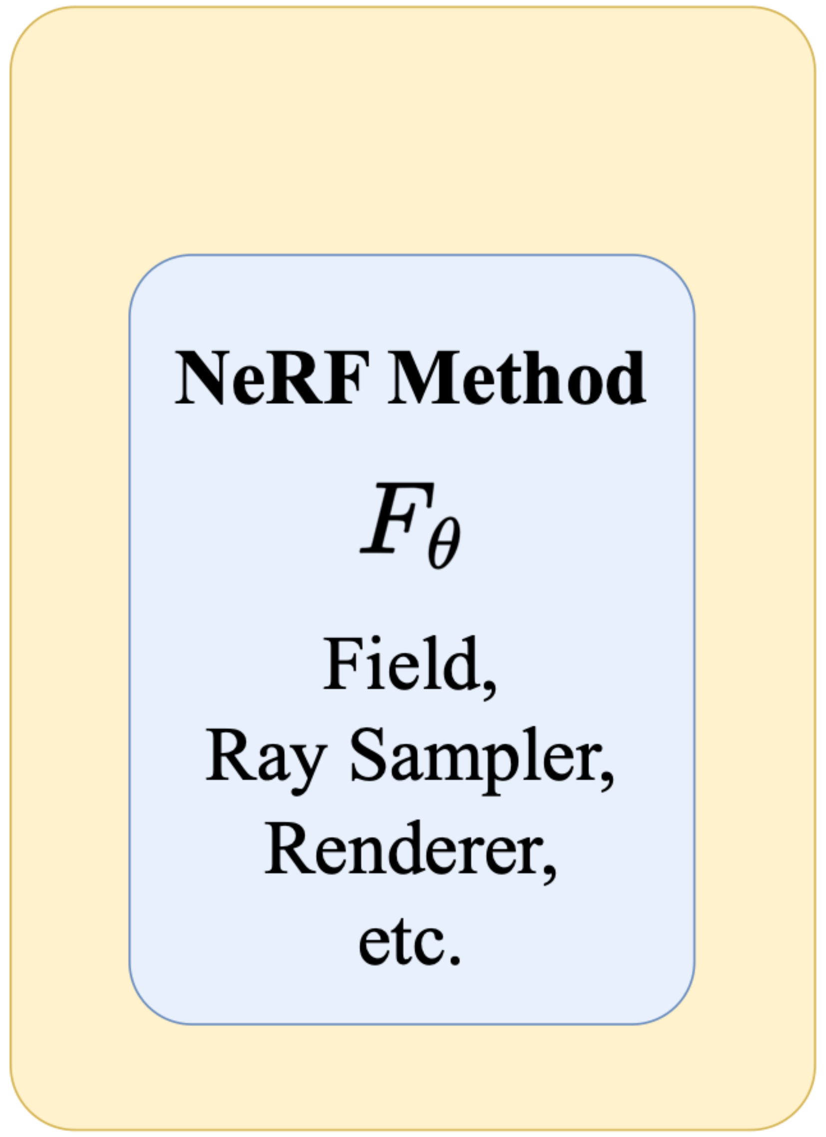
Goal

- **Propose a latent NeRF training technique compatible with most NeRF architectures**
- Build a 3D-aware latent space in which training NeRF is possible

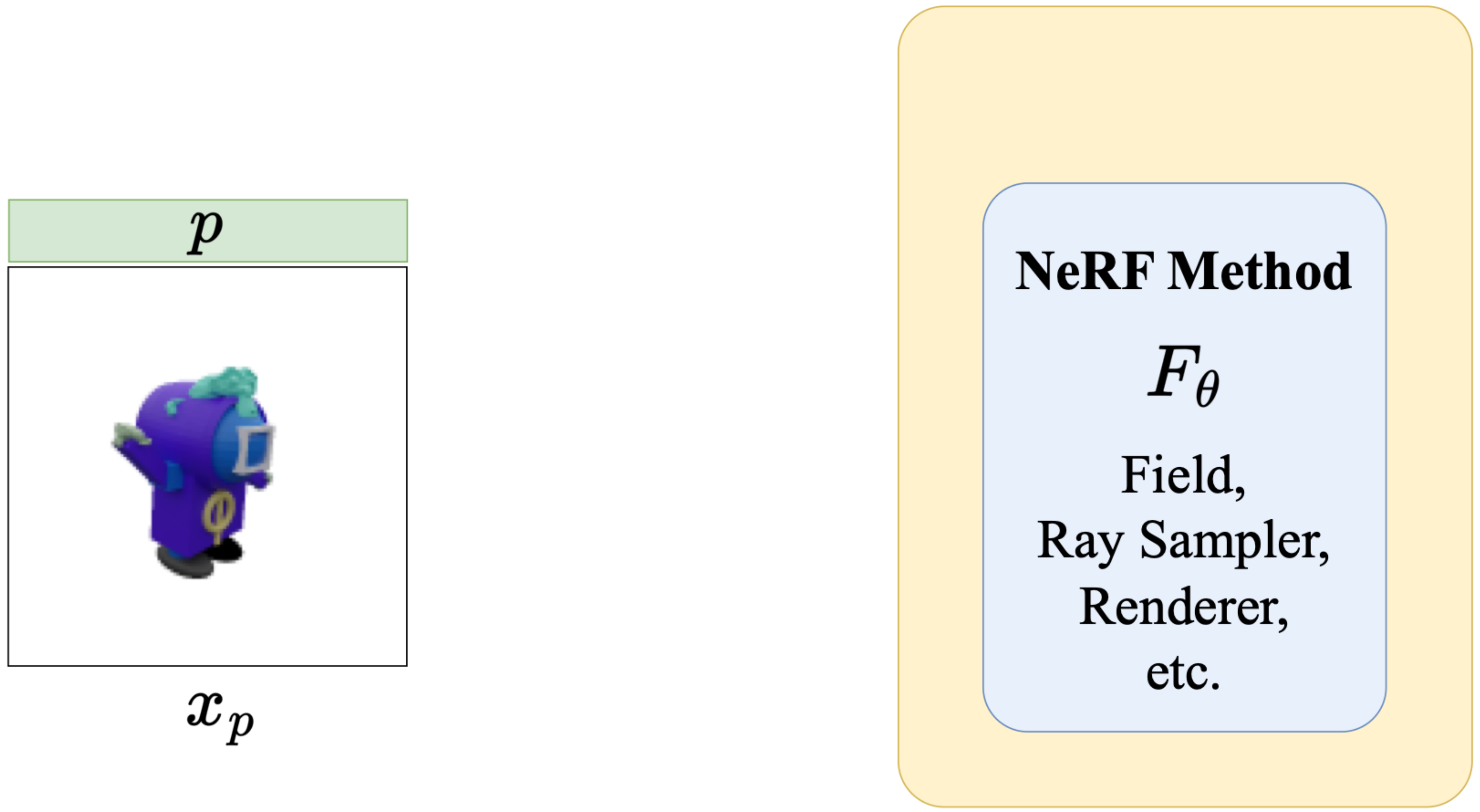


(a) Image space analogy.

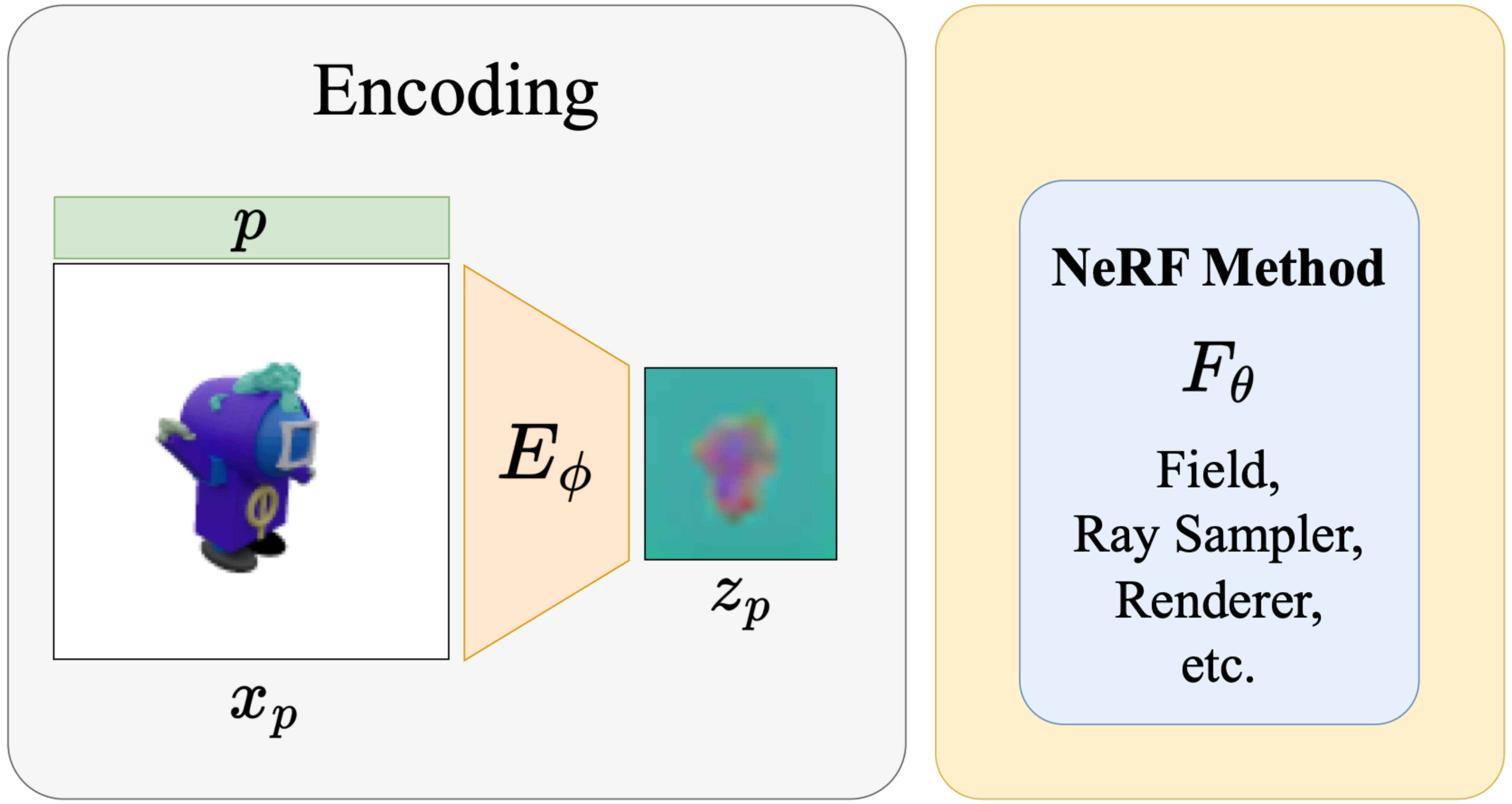
Latent NeRF Training Pipeline



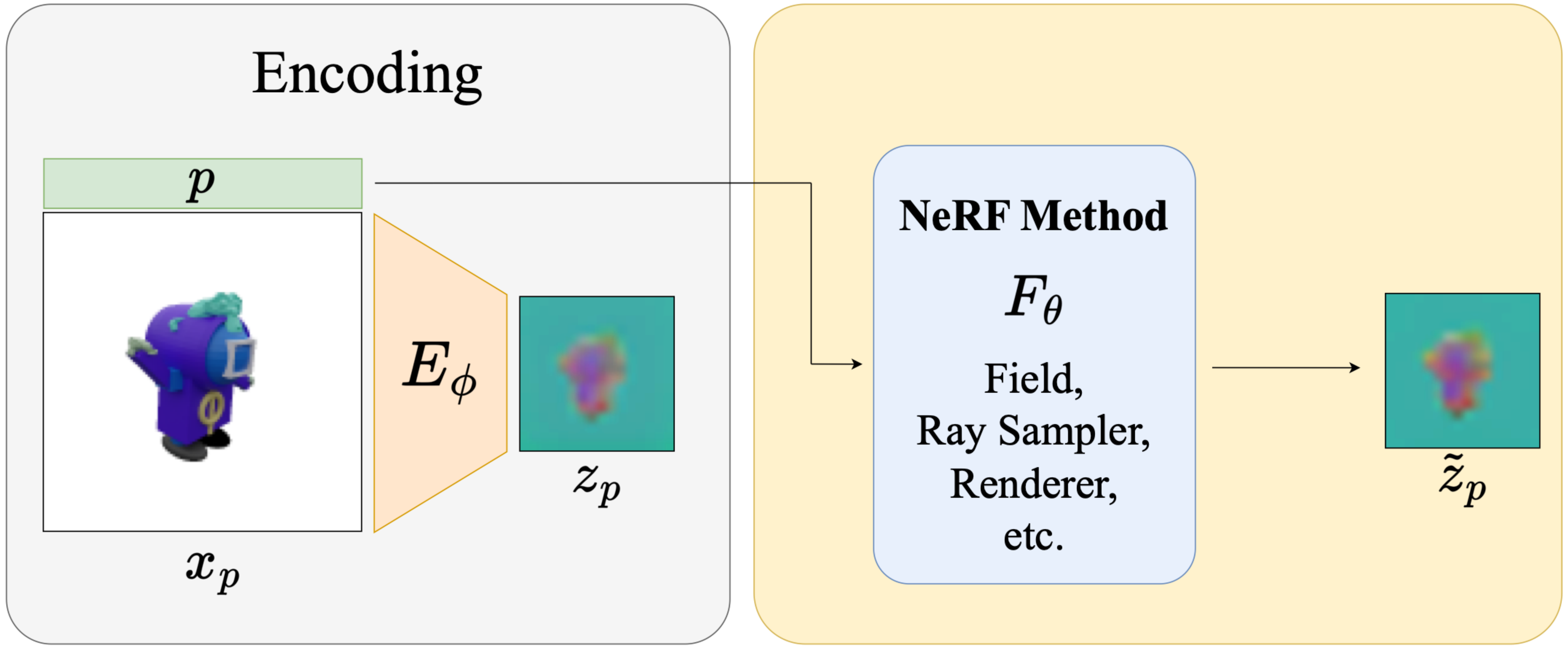
Latent NeRF Training Pipeline



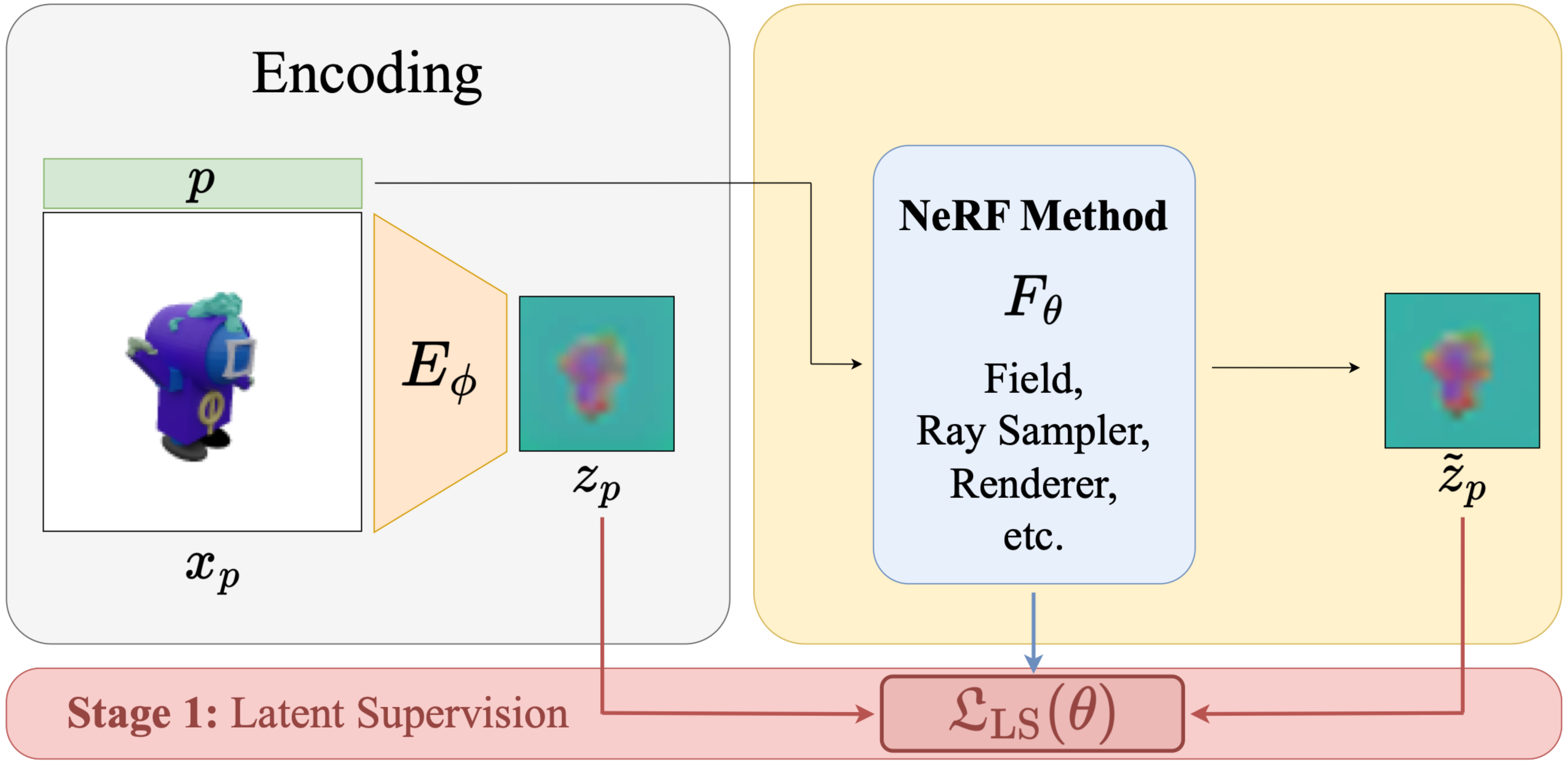
Latent NeRF Training Pipeline



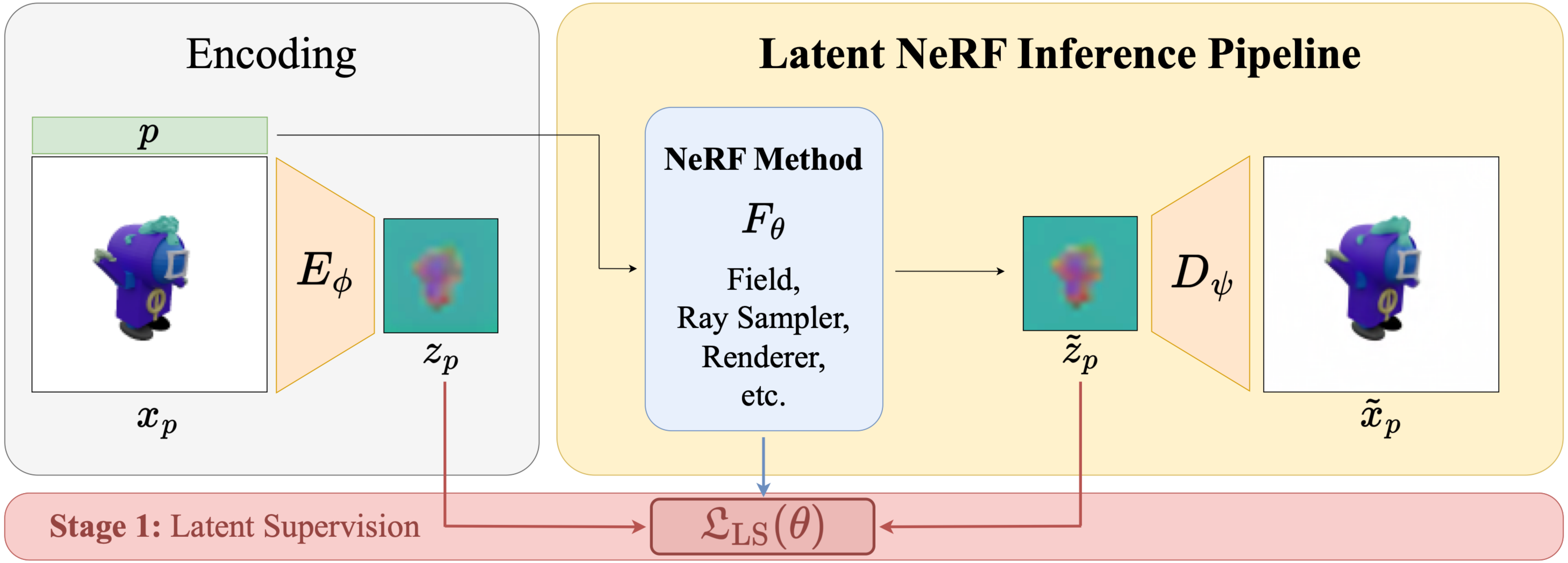
Latent NeRF Training Pipeline



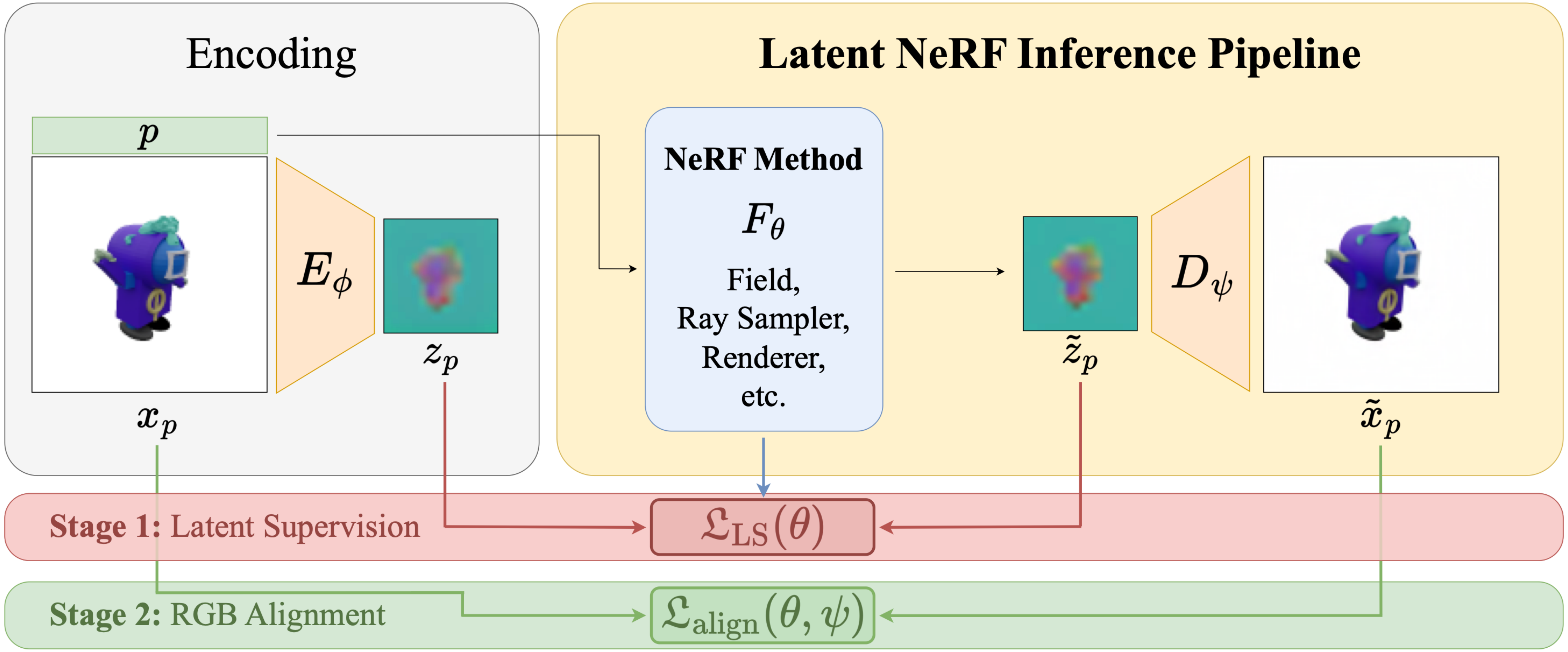
Latent NeRF Training Pipeline



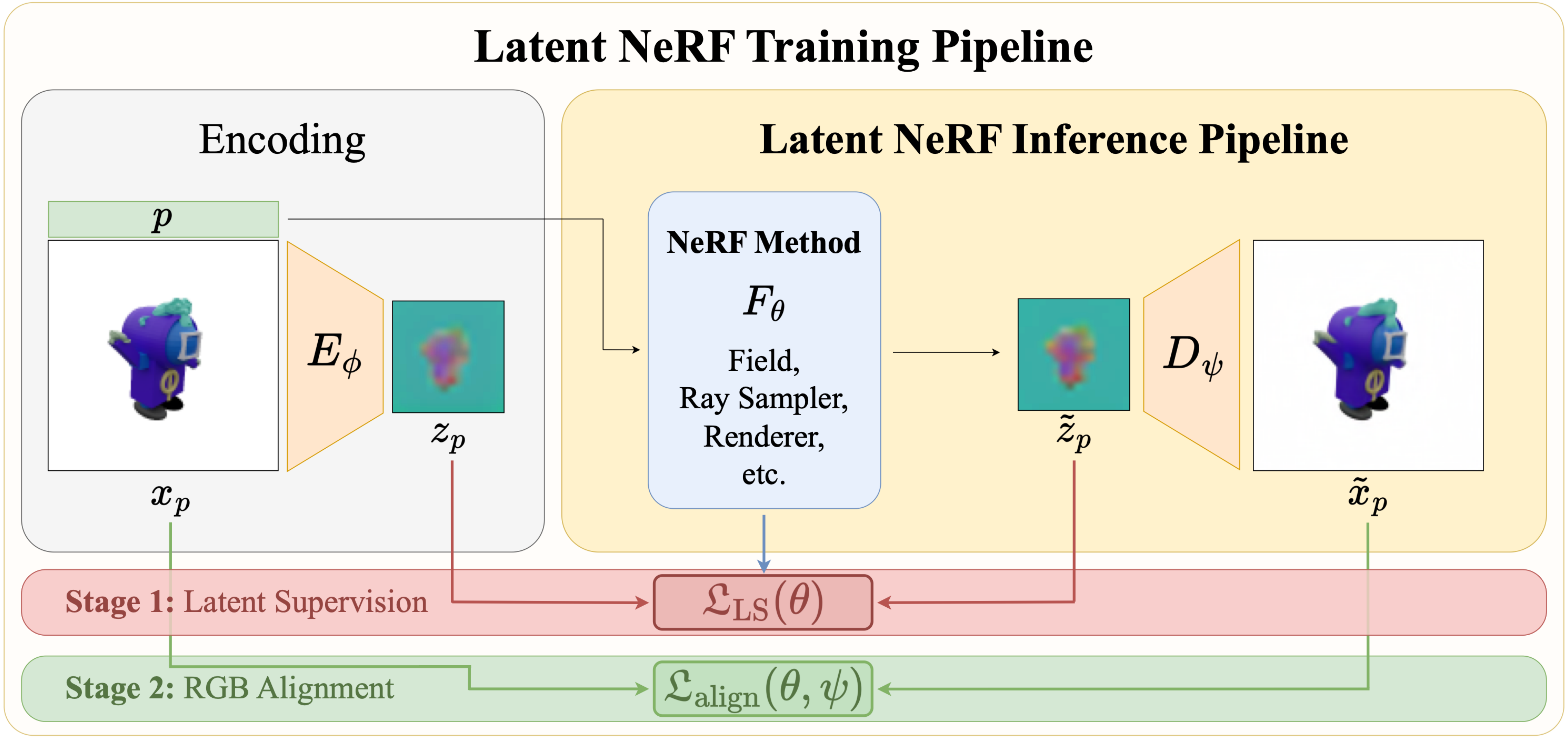
Latent NeRF Training Pipeline



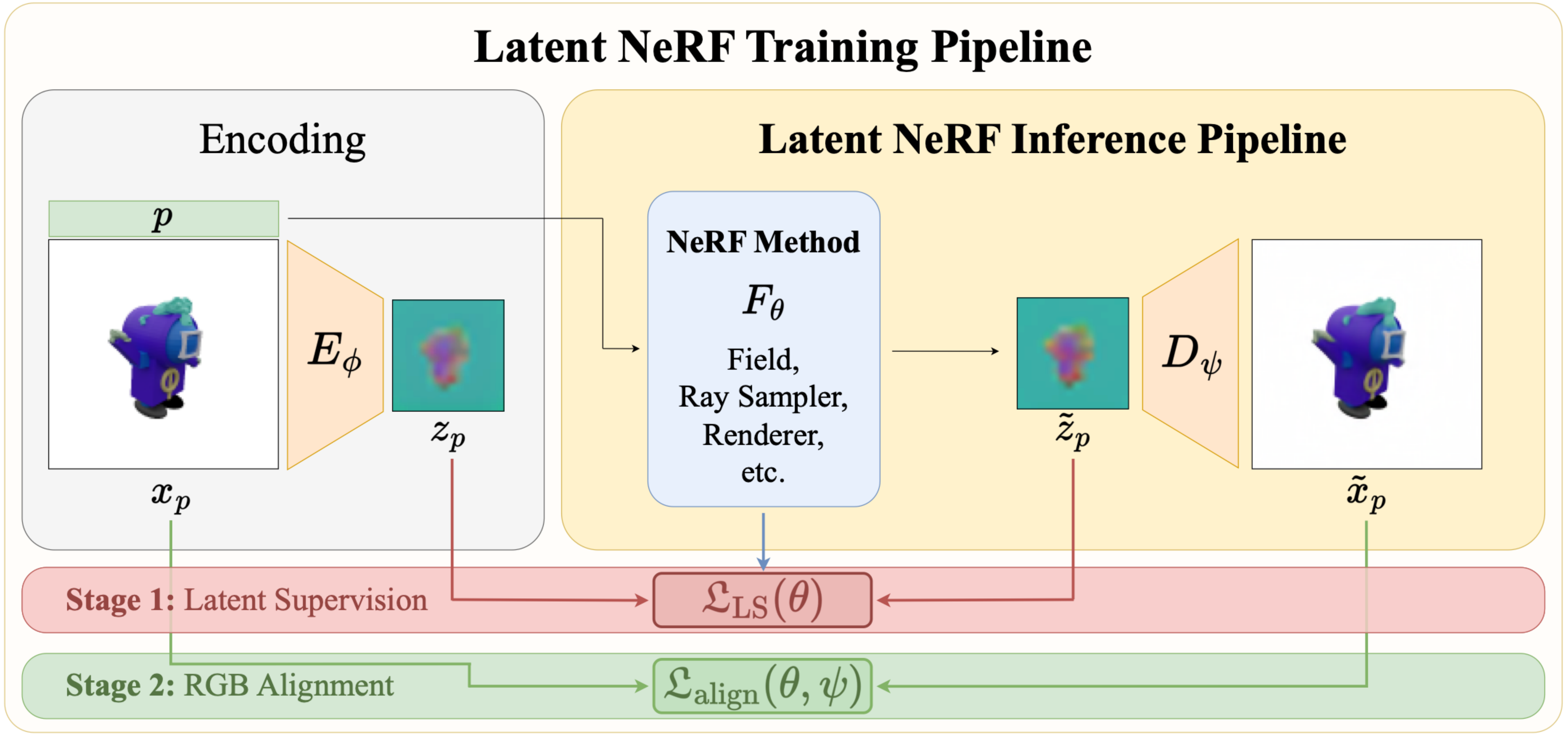
Latent NeRF Training Pipeline

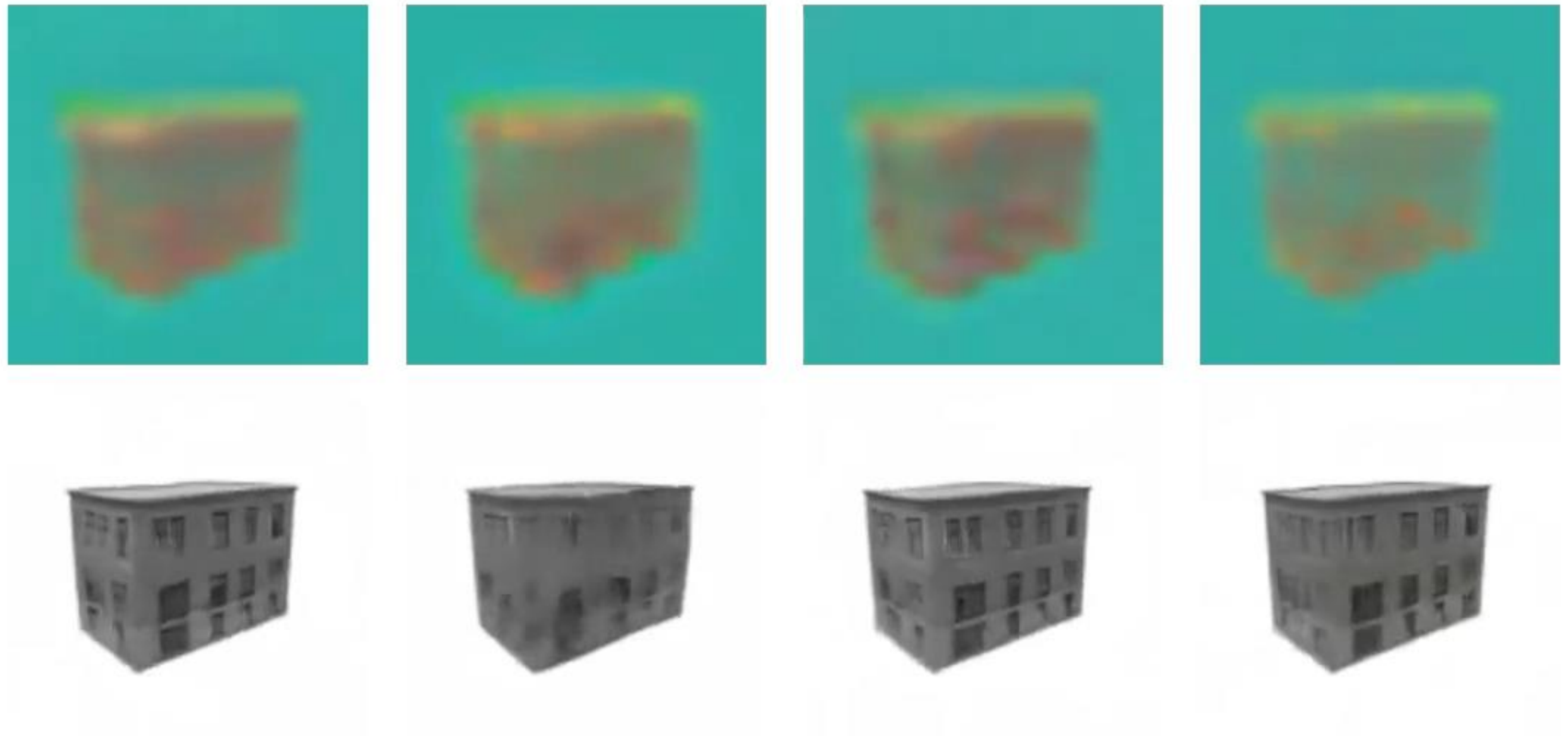


Latent NeRF Training Pipeline



Latent NeRF Training Pipeline



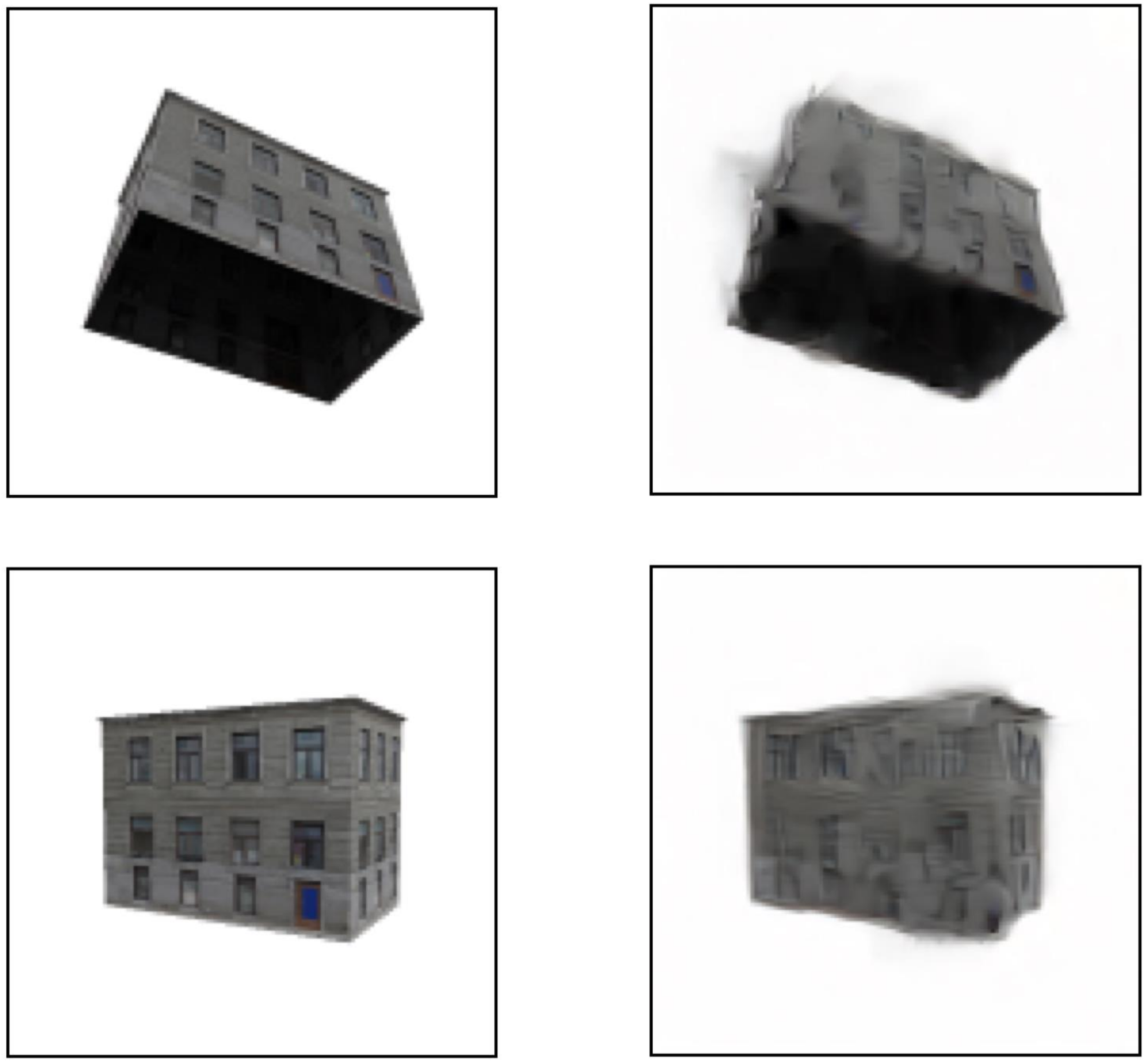


Vanilla-NeRF | Instant-NGP | TensoRF | K-Planes

open-source
nerfstudio
extension

ig-ae.github.io

Results on a standard AE



Ground Truth

Decoded
NeRF Rendering

Goal

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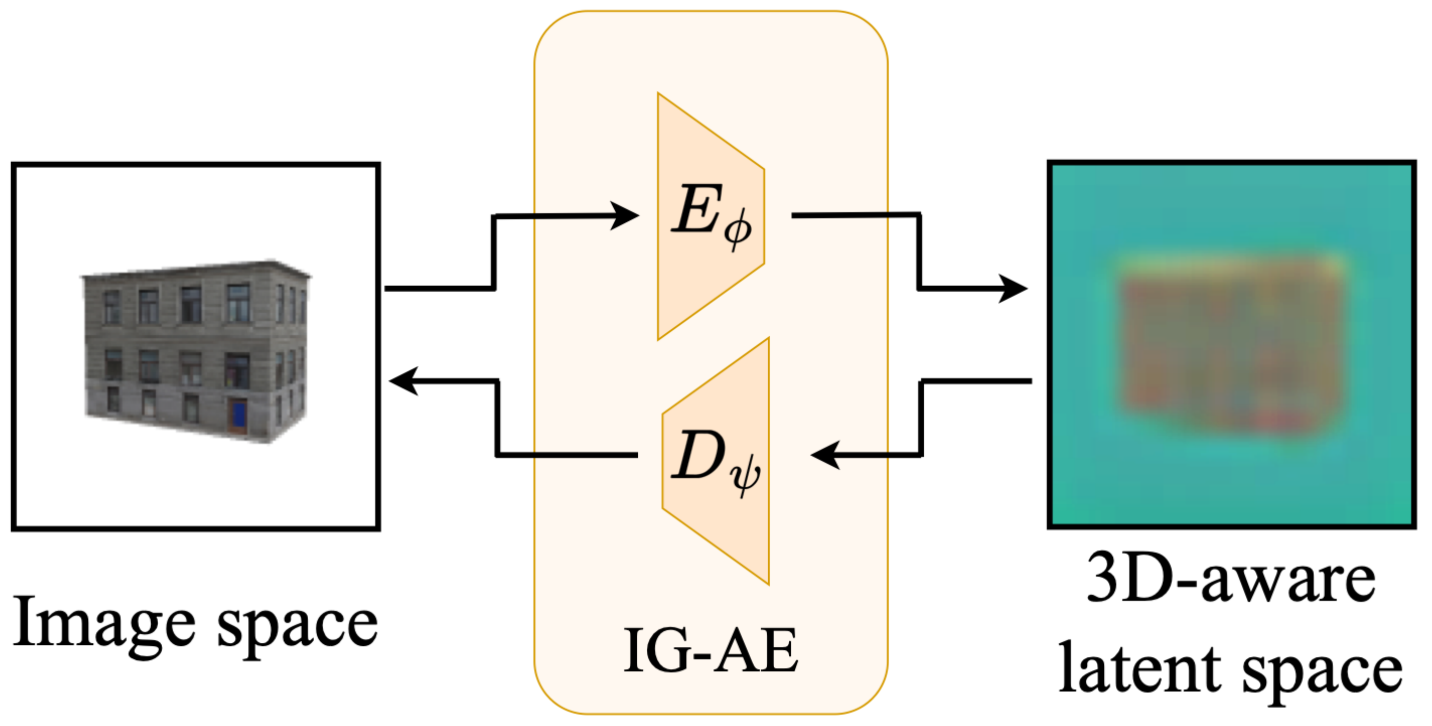
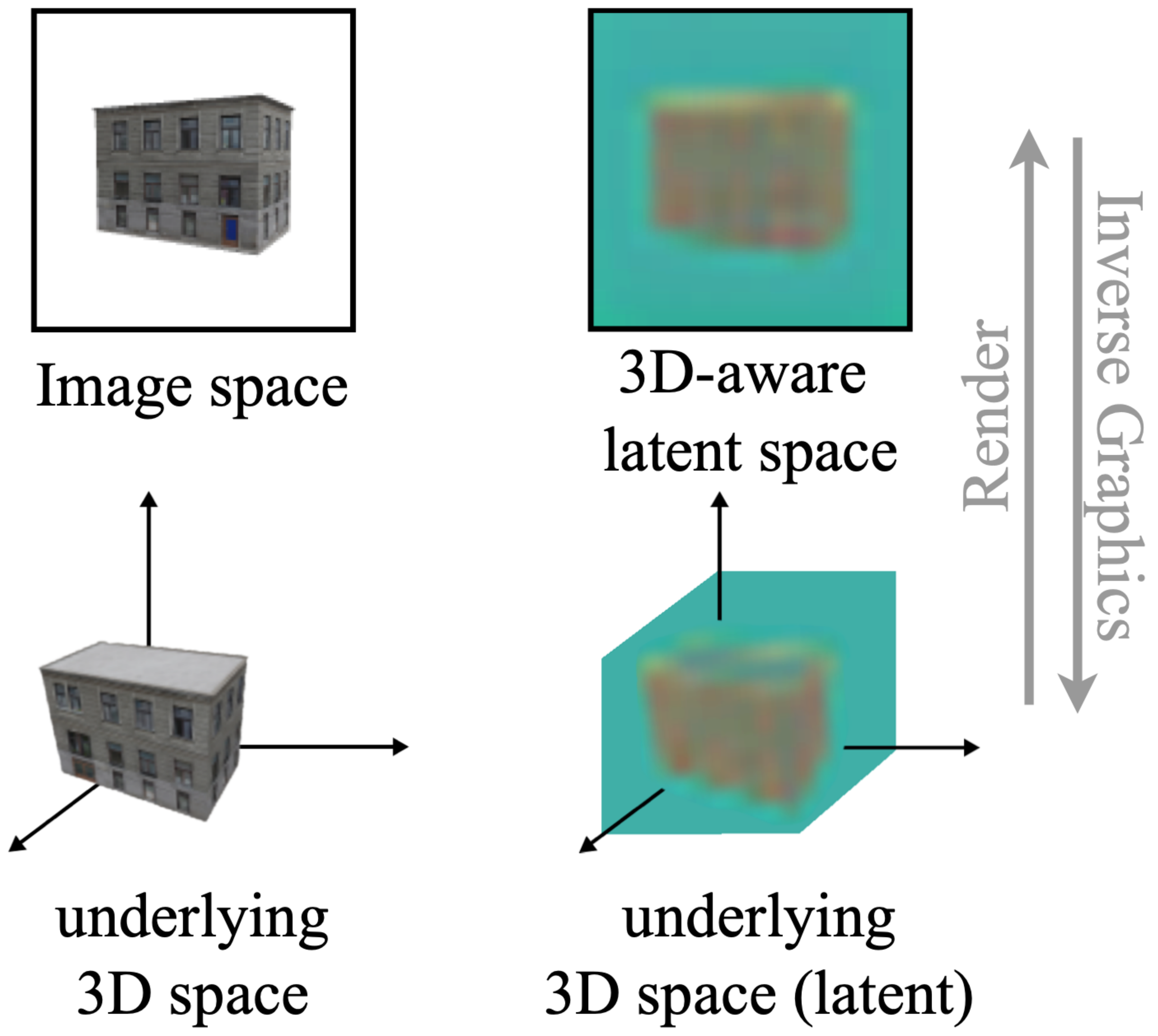
- **Build a 3D-aware latent space in which training NeRF is possible**

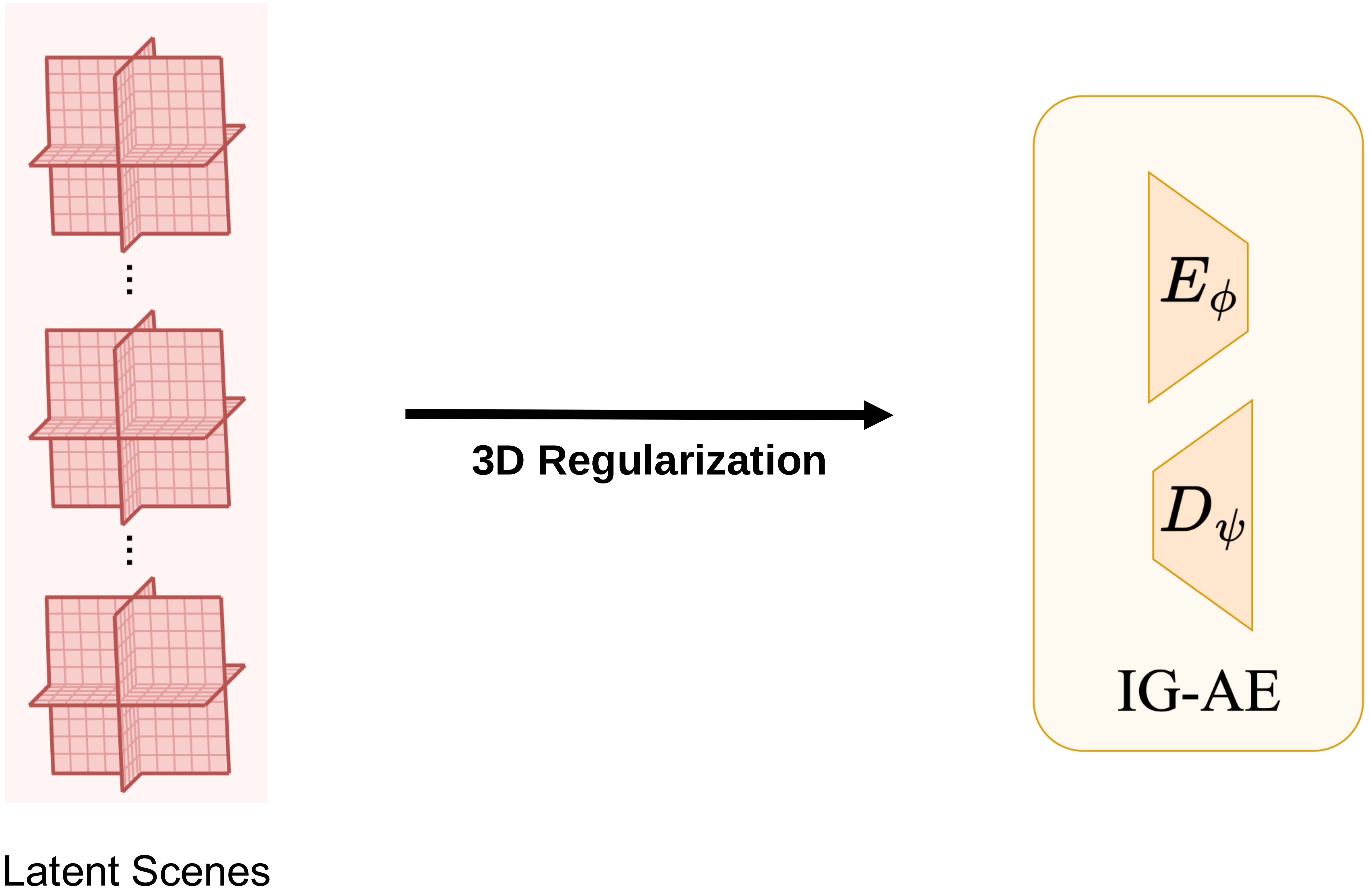


Inverse Graphics Autoencoder

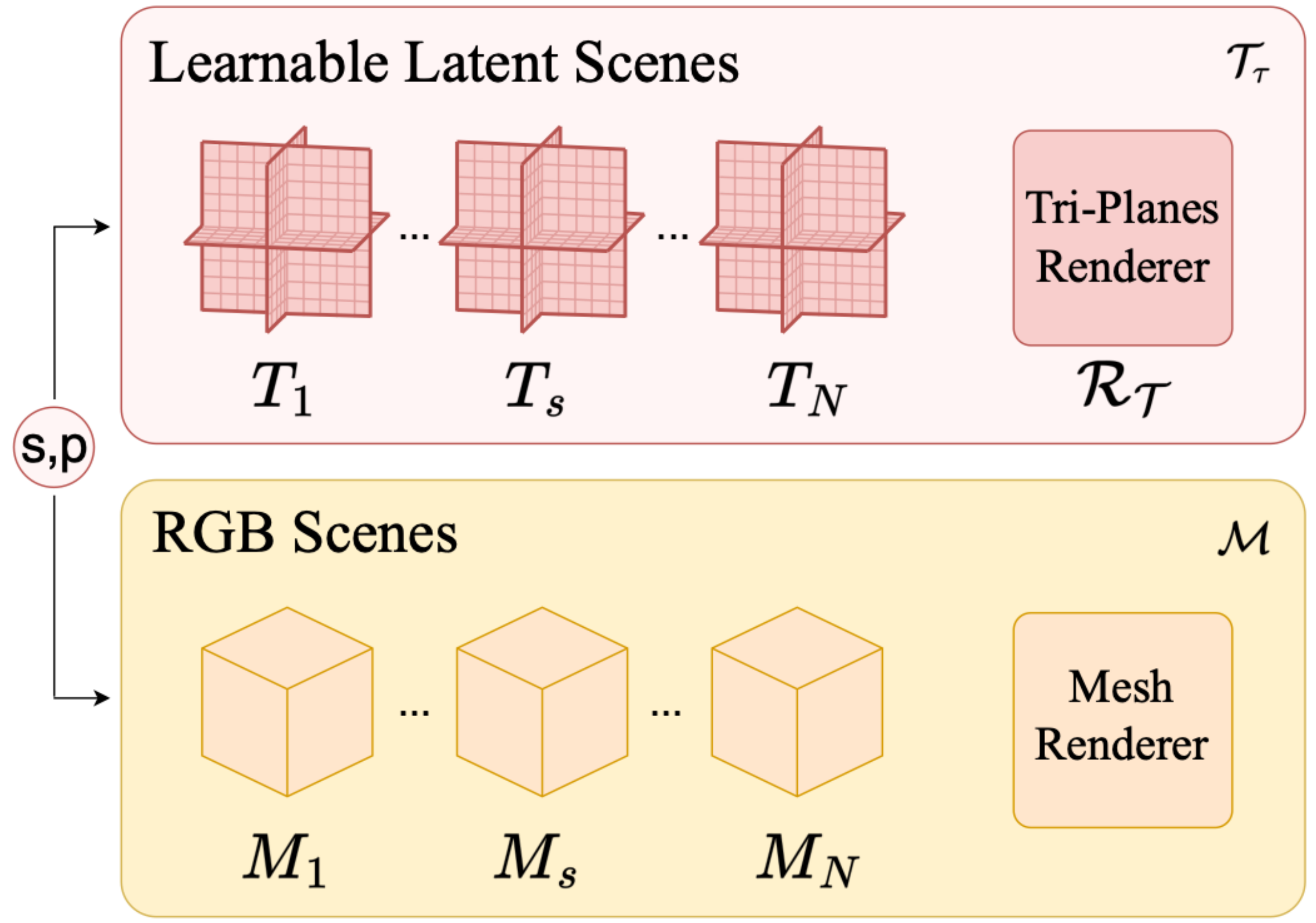


(a) Image space analogy.

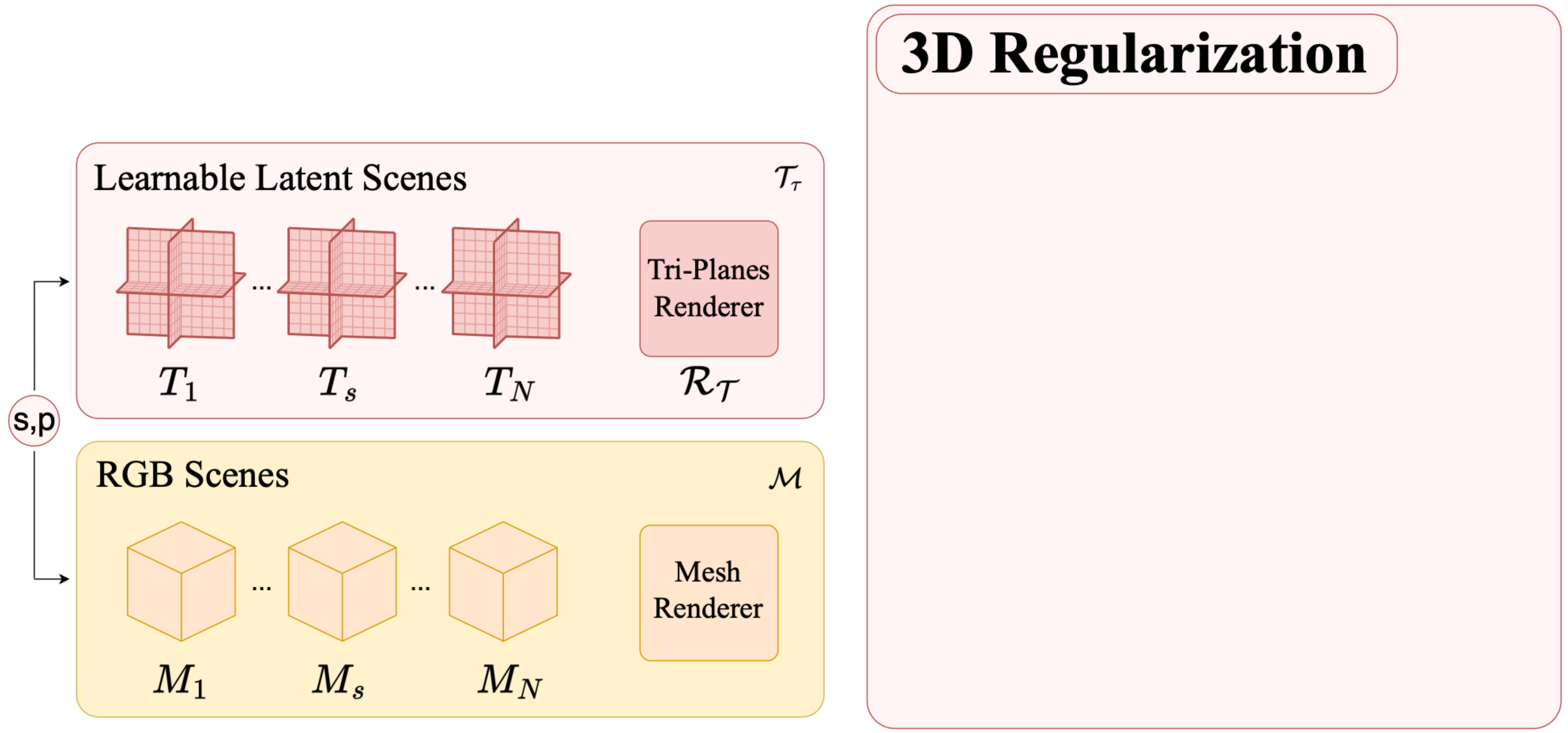
IG-AE Training Pipeline



IG-AE Training Pipeline



IG-AE Training Pipeline



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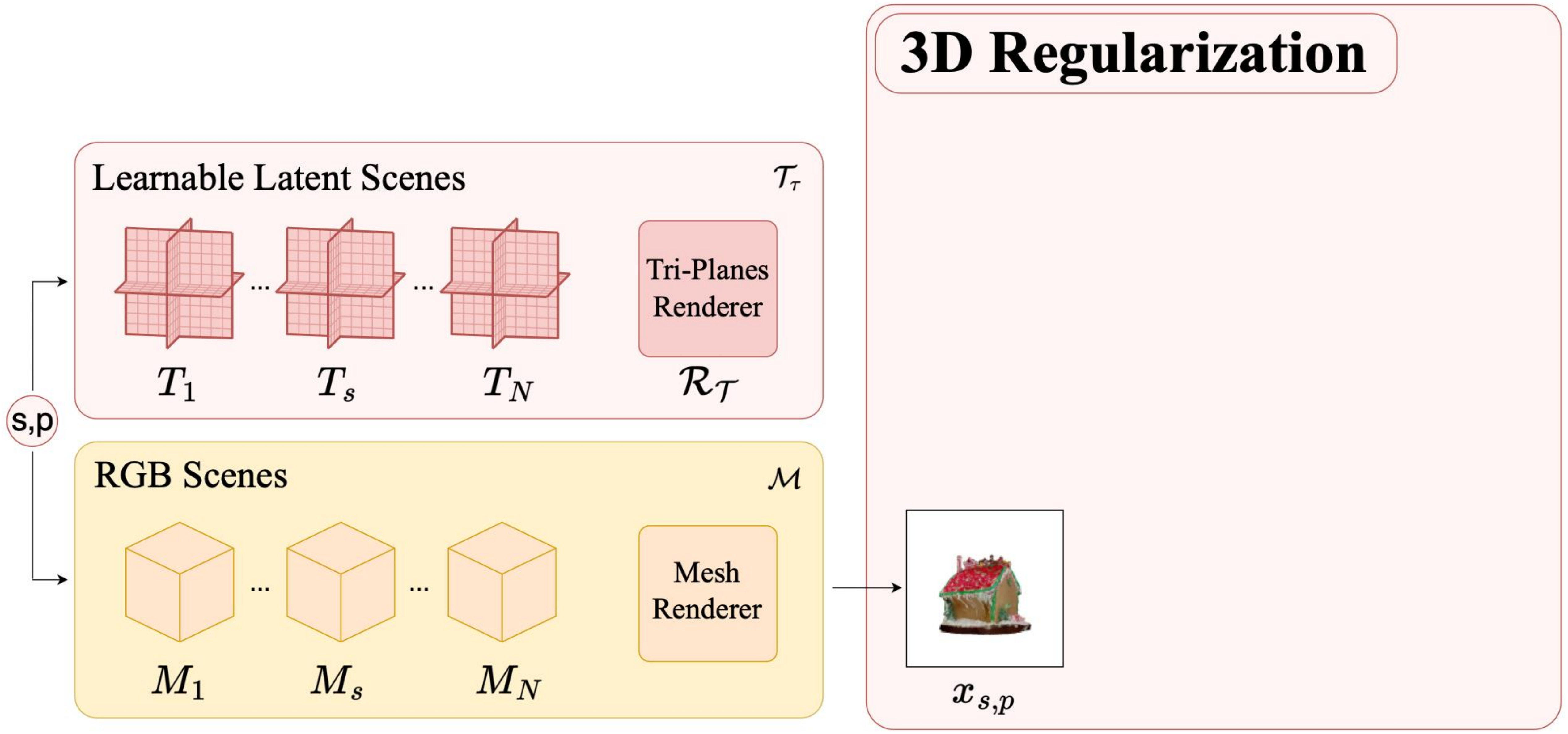

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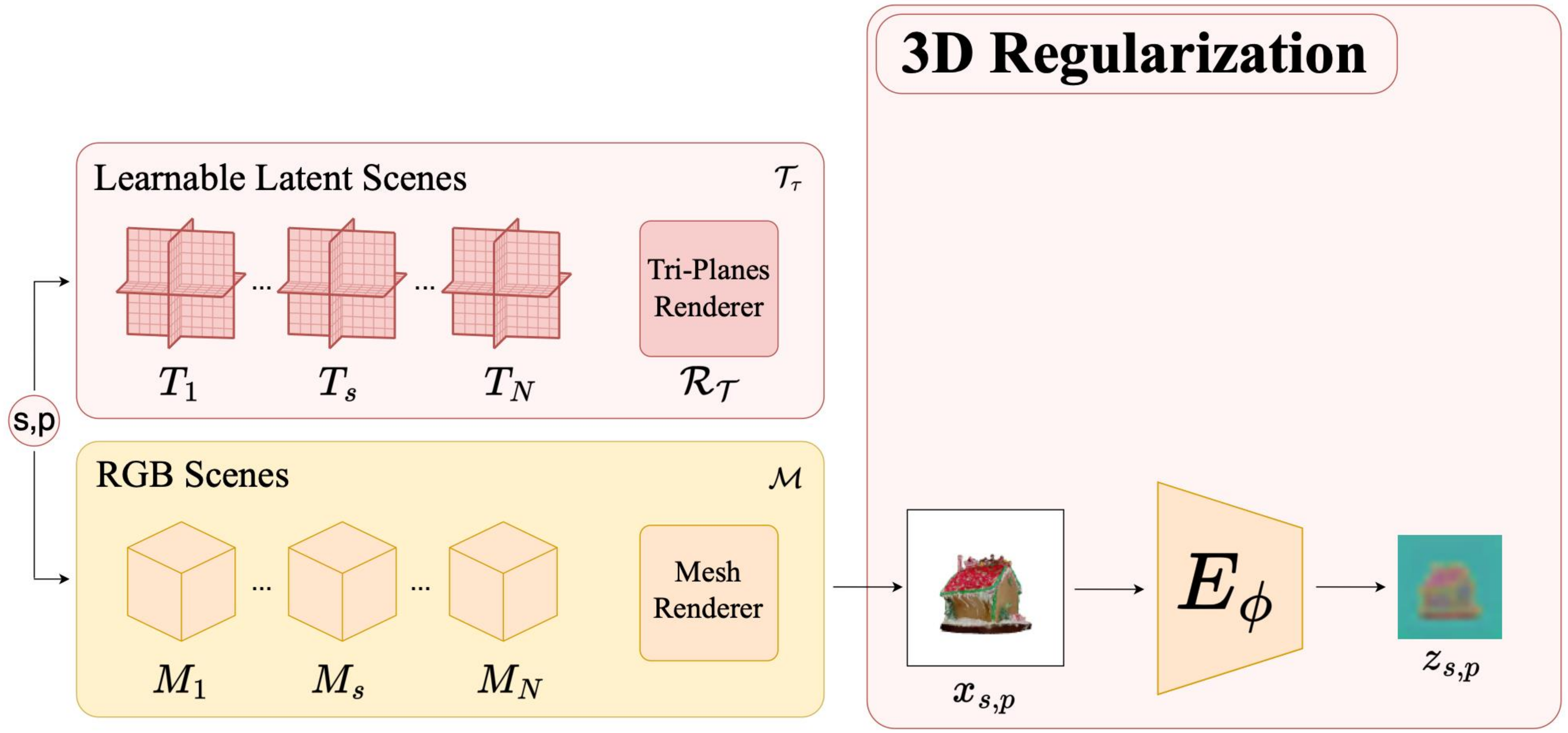

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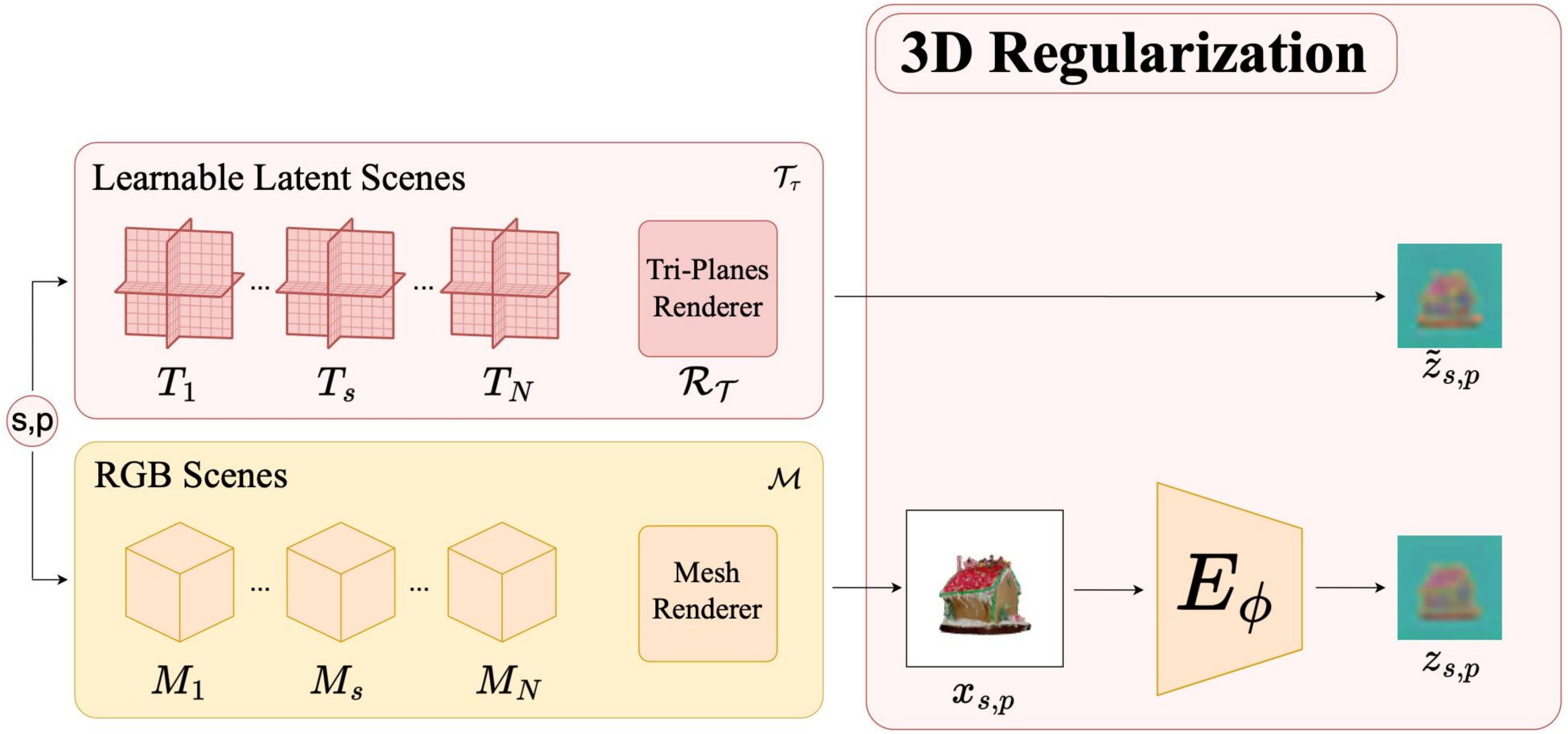
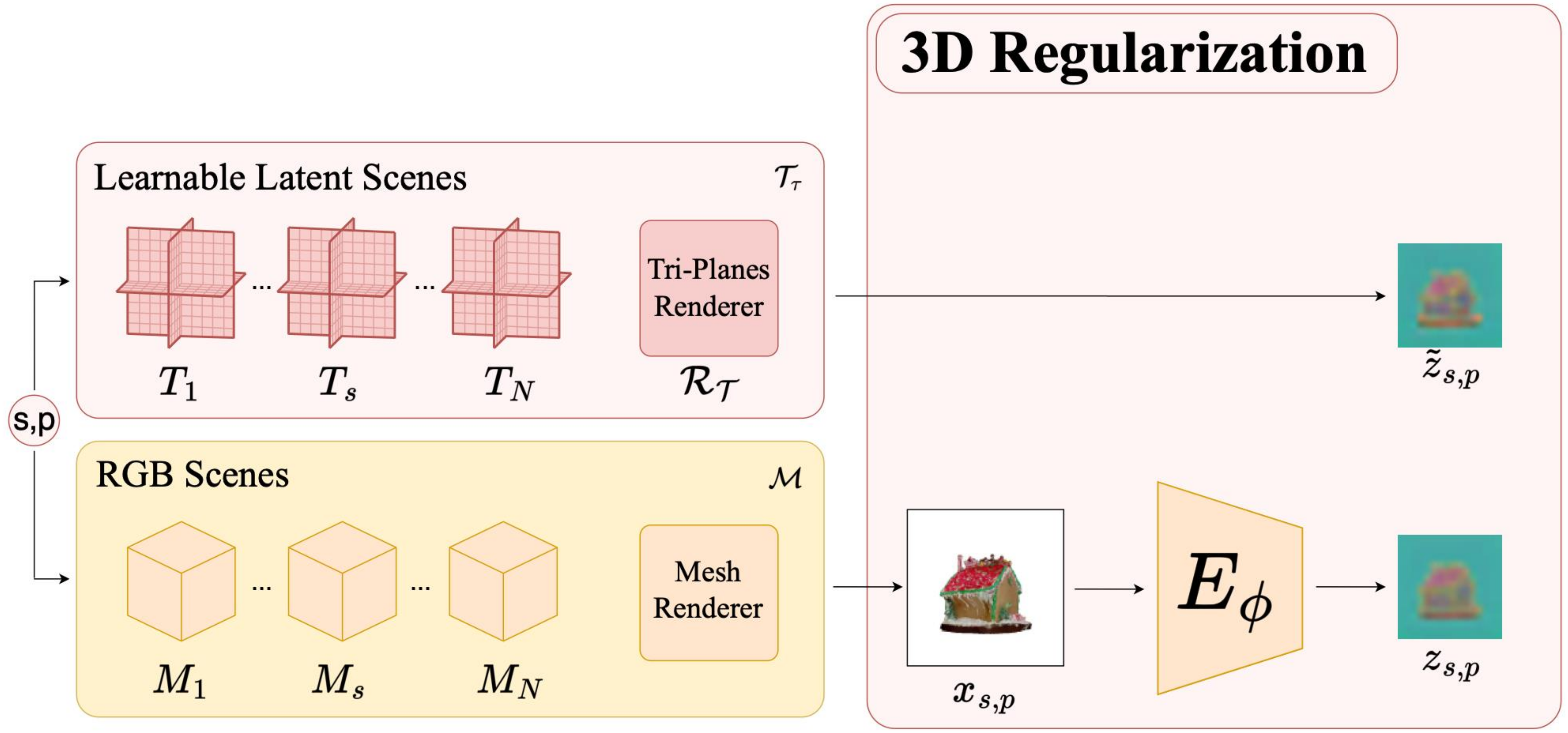


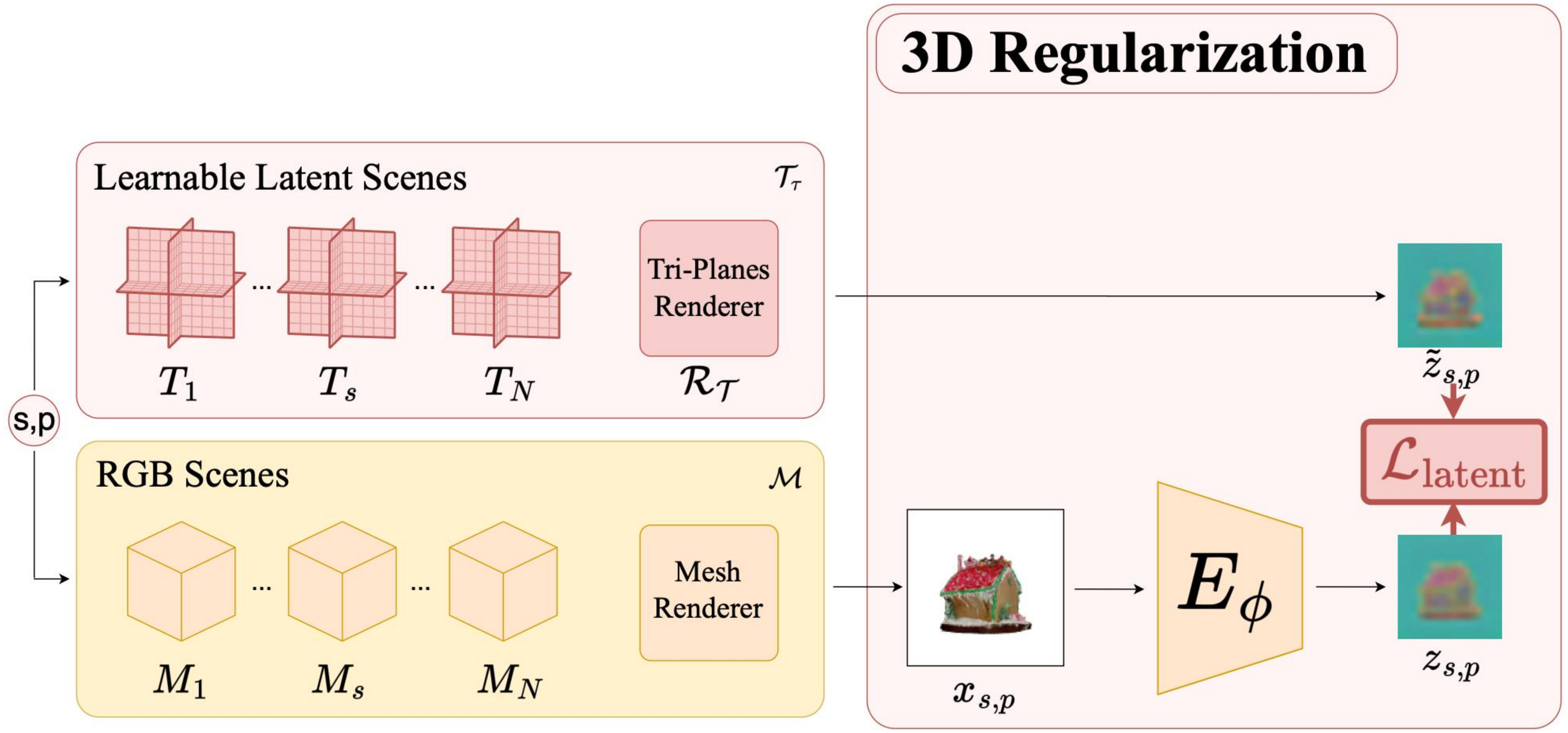
IG-AE Training Pipeline



😄 3D consistent

🙁 non-3D consistent

IG-AE Training Pipeline

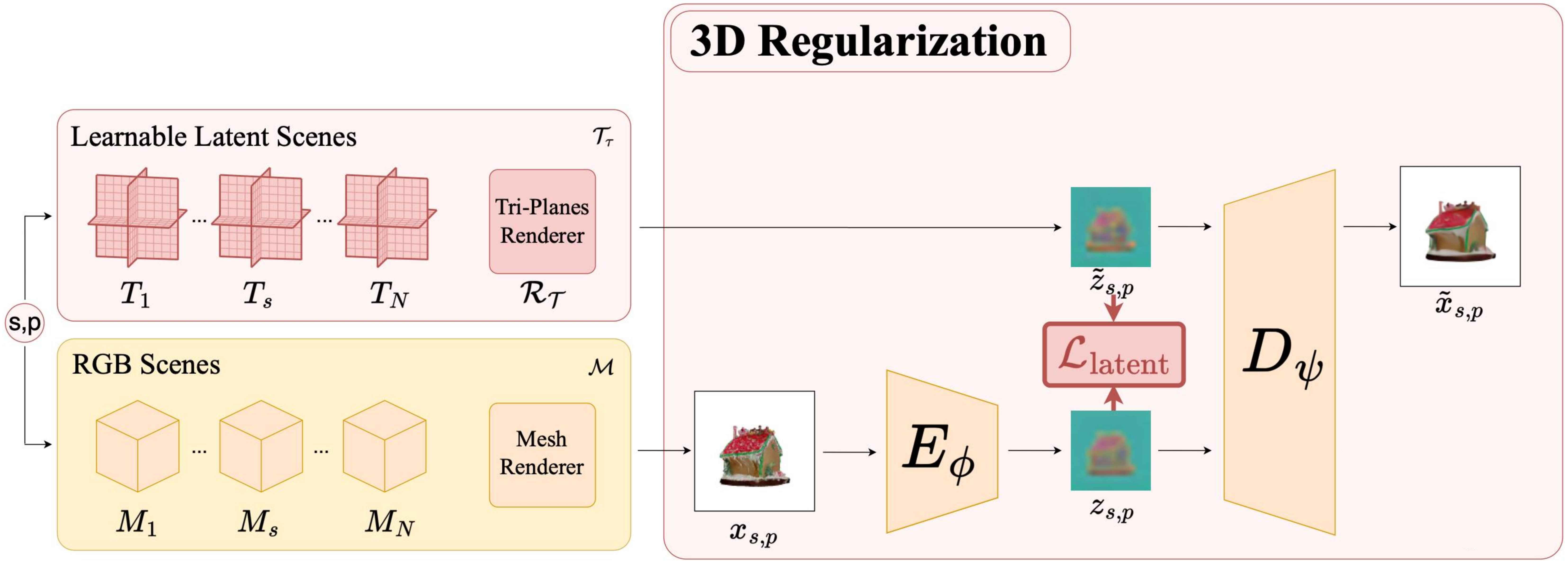


😄 3D consistent

🙁 non-3D consistent

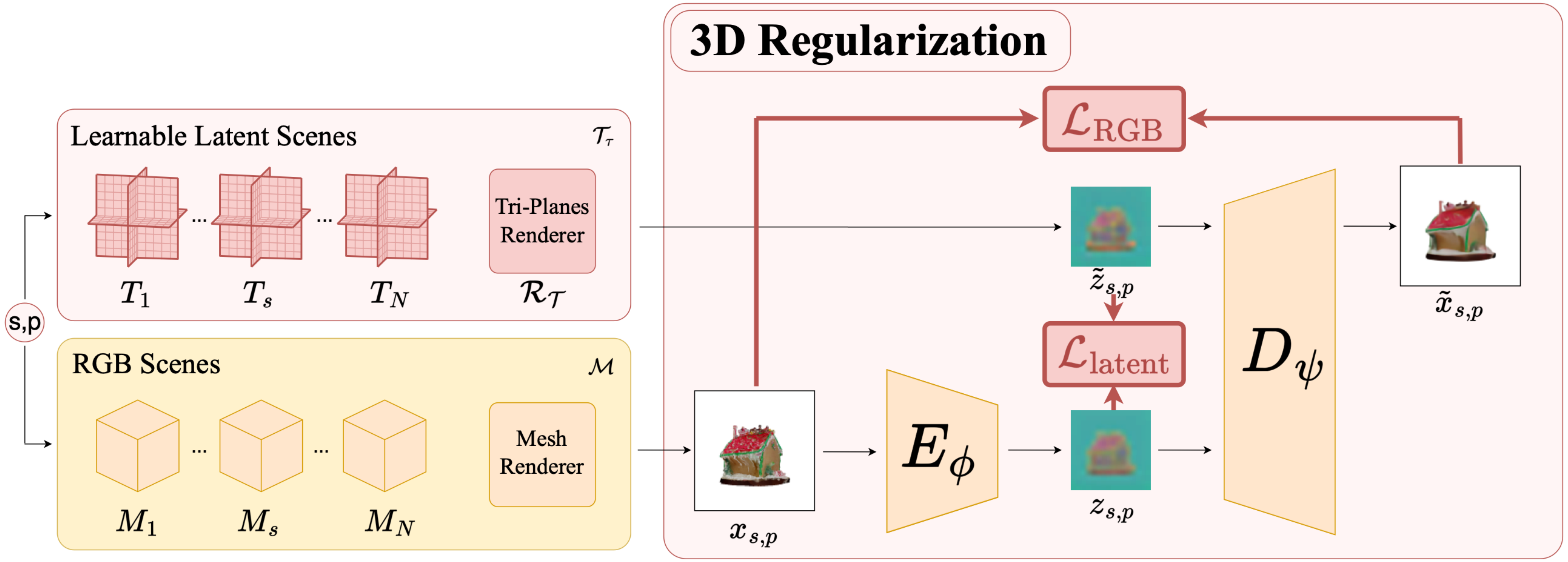
✅ **3D consistent Encoder**

IG-AE Training Pipeline



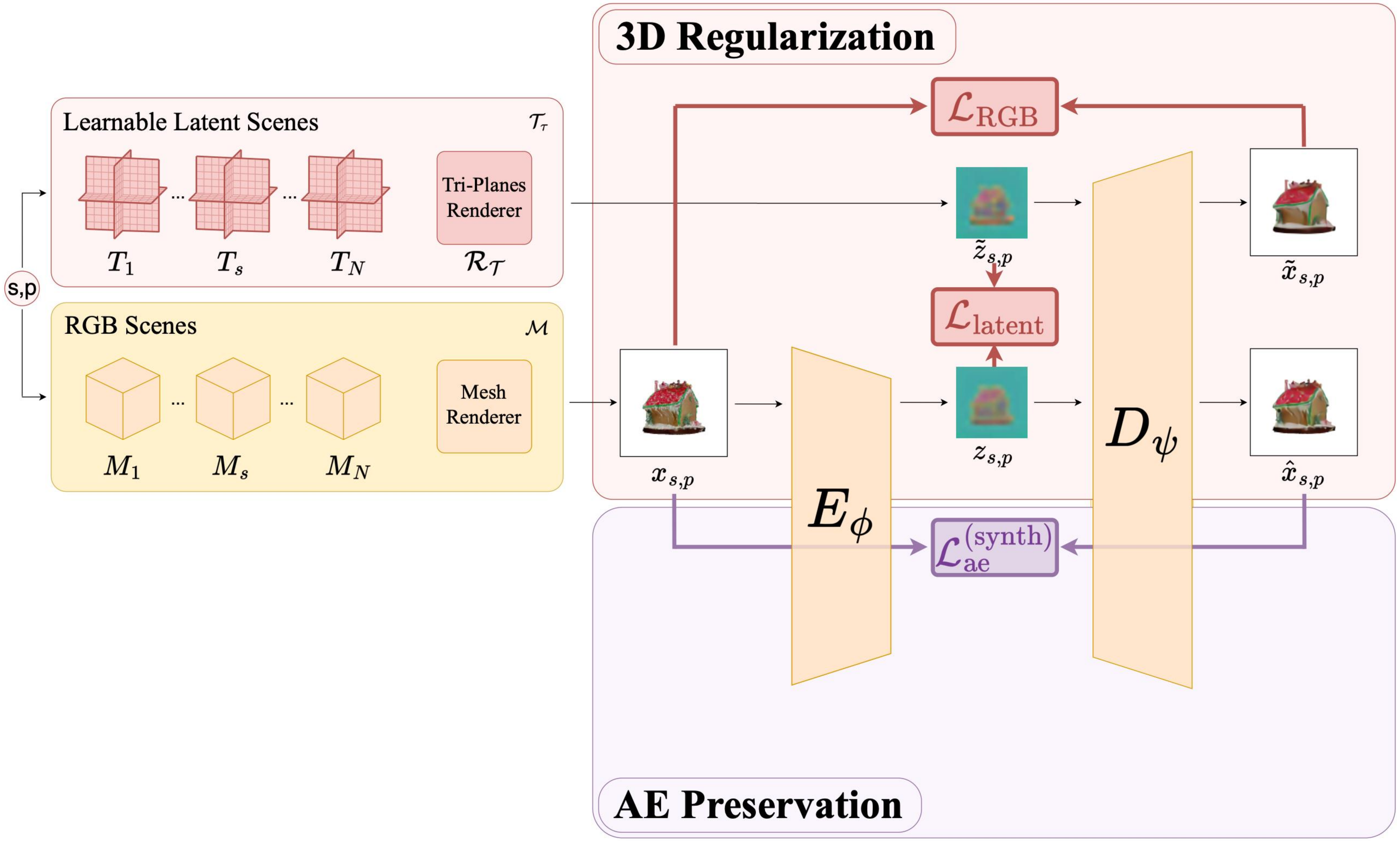
✅ **3D consistent Encoder**

IG-AE Training Pipeline

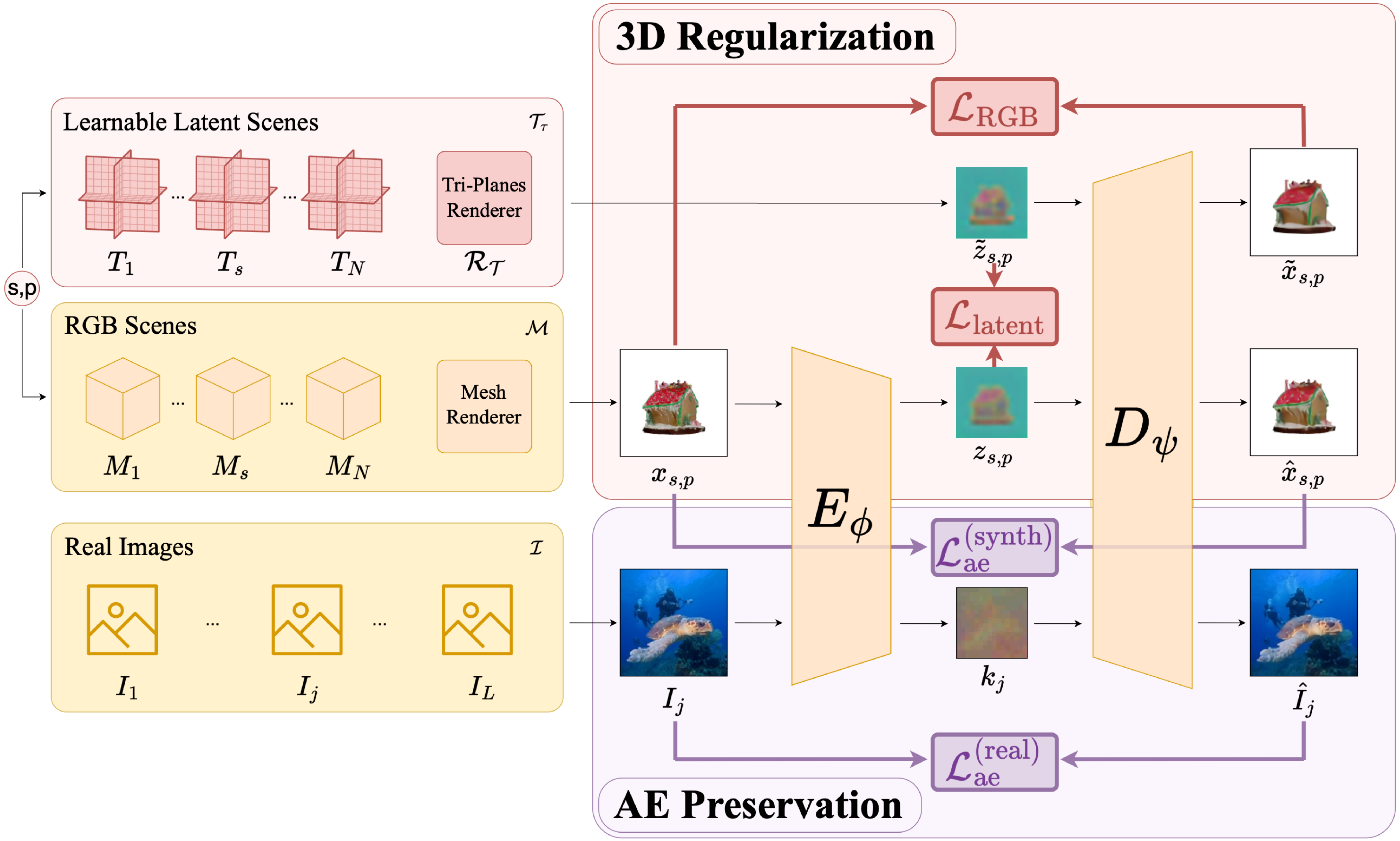


✅ 3D consistent Encoder
✅ 3D consistent Decoder

IG-AE Training Pipeline



IG-AE Training Pipeline



Results

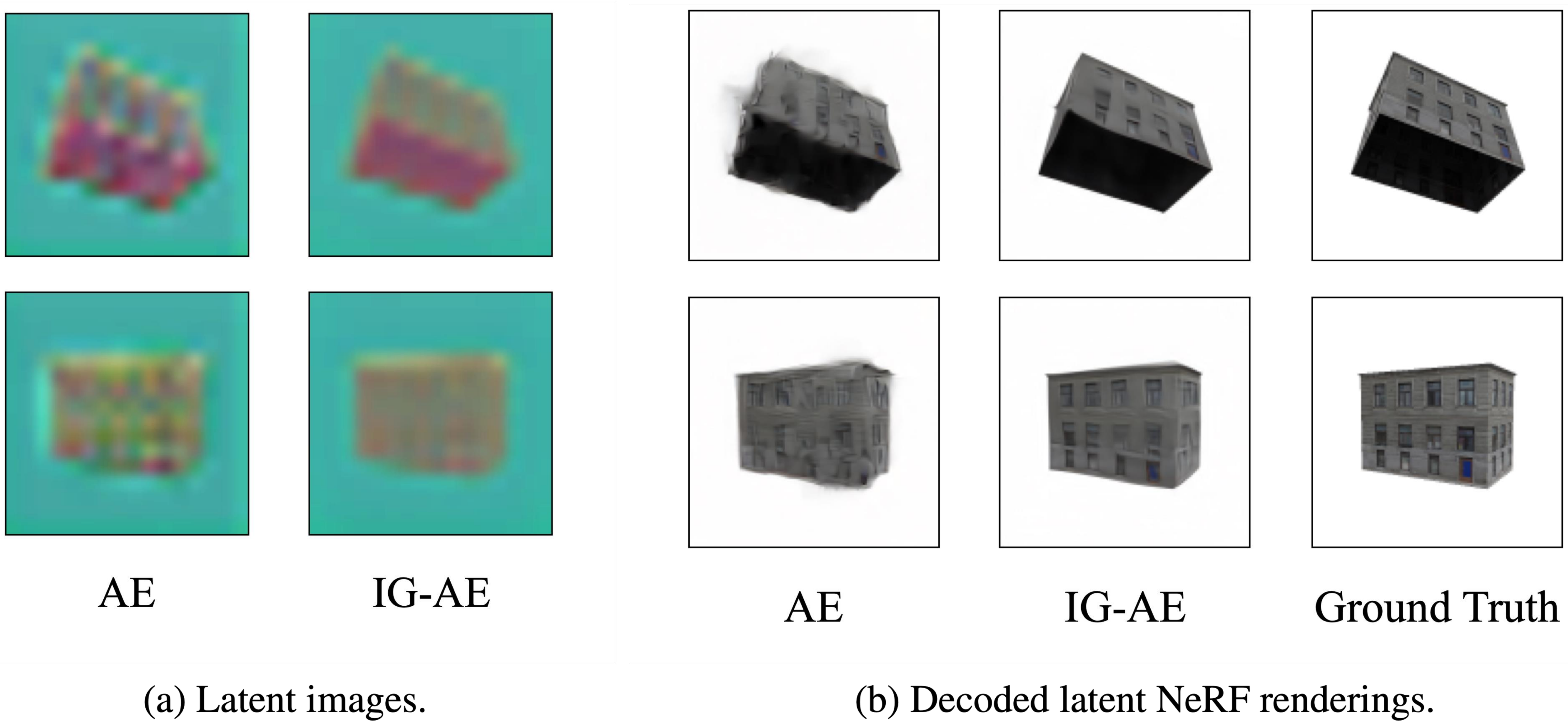
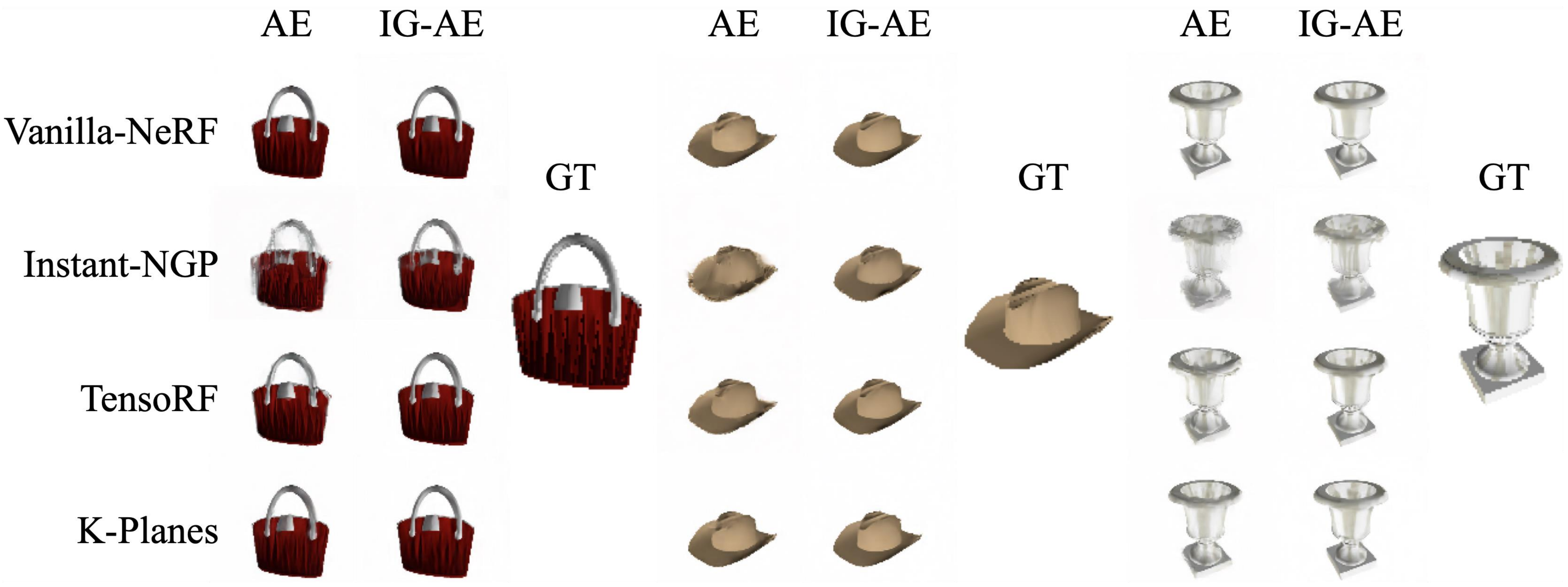


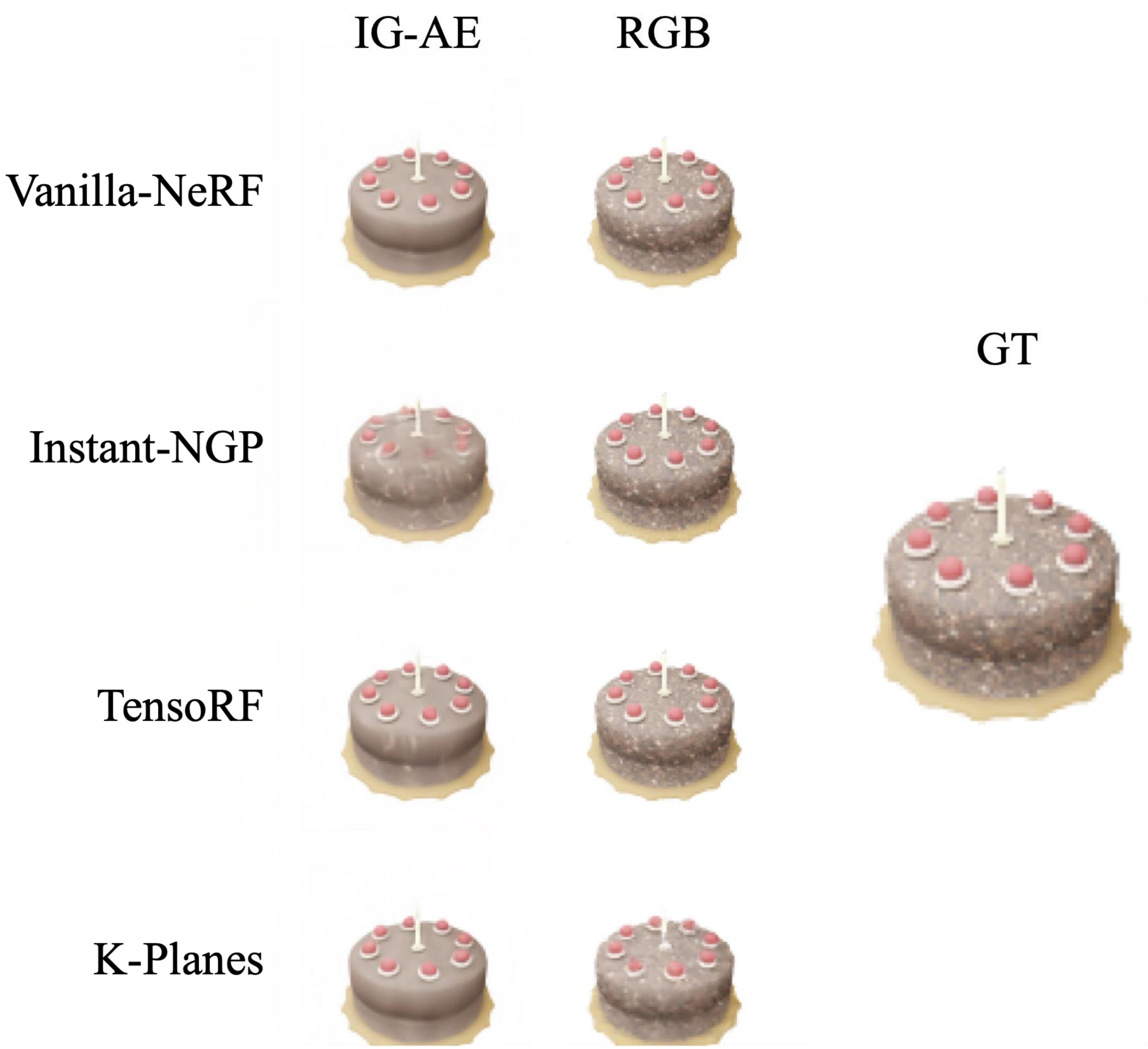
Figure 2: **Comparison of IG-AE and a standard AE.** Encoding 3D-consistent images using an AE leads to 3D-inconsistent latent images. When trained on such latents, NeRF renderings present artifacts when decoded. IG-AE presents a 3D-aware latent space with 3D-consistent latent images. Latent NeRFs trained with our IG-AE omit these artifacts and more closely match the ground truth.

Results



Comparison of Latent NeRFs trained with AE vs IG-AE

Comparison with RGB NeRF



Rendering Quality

	Training Space	Training Time (min)	Rendering Time (ms)
Vanilla NeRF	RGB IG-AE	637 28	1201 19.7
Instant NGP	RGB IG-AE	6 19	11.3 7.3
TensoRF	RGB IG-AE	92 20	101.4 10.4
K-Planes	RGB IG-AE	88 29	63.3 11.3

Training and Rendering Times

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